



Clonal Diversity at Early Cancer Recurrence

Kevin Leder¹ · Zicheng Wang² 

Received: 10 December 2025 / Accepted: 11 February 2026 / Published online: 6 April 2026
© The Author(s), under exclusive licence to the Society for Mathematical Biology 2026

Abstract

Despite initial success, cancer therapies often fail due to the emergence of drug-resistant cells. In this study, we use a mathematical model to investigate how cancer evolves over time, specifically focusing on the state of the tumor when it recurs after treatment. We use a two-type birth-death process to capture the dynamics of both drug-sensitive and drug-resistant cells. Assuming resistant cells have equal fitness, we analyze the clonal diversity of drug-resistant cells at the time of cancer recurrence, which is defined as the first time the population size of drug-resistant cells exceeds a specified proportion of the initial population size of drug-sensitive cells. We examine two clonal diversity indices: the number of clones and the Simpson's Index. We calculate the expected values of these indices at the time of cancer recurrence. Additionally, we examine these two indices conditioned on early recurrence in the special case of a deterministically decaying sensitive population, with the aim of addressing the question of whether early recurrence is driven by a single mutation that generates an unusually large family of drug-resistant cells (corresponding to a low clonal diversity), or if it is due to the presence of an unusually large number of mutations causing drug resistance (corresponding to a high clonal diversity). Our findings, based on both indices, support the latter possibility. Furthermore, we demonstrate that the time of cancer recurrence can serve as a valuable indicator of clonal diversity, providing new insights into the evolutionary dynamics of recurrent cancers.

Keywords Cancer recurrence · Clonal diversity · Birth–death process

1 Introduction

Many cancers possess a remarkable ability to develop drug resistance through genetic mutations that occur during treatments, a phenomenon referred to as acquired resis-

✉ Zicheng Wang
wangzicheng@cuhk.edu.cn

Kevin Leder
lede0024@umn.edu

¹ Department of Industrial and Systems Engineering, University of Minnesota, Twin Cities, US

² School of Data Science, The Chinese University of Hong Kong, Shenzhen, China

tance¹. Acquired resistance may result from the activation of a second proto-oncogene that emerges as a new driver gene or from mutations in the drug targets themselves (Wang et al. 2019). This adaptability poses a significant challenge to the effectiveness of cancer treatments and can lead to cancer recurrence. One example of acquired resistance is observed in colorectal cancer, where amplification or mutations of the KRAS gene can confer resistance to anti-EGFR therapies such as Cetuximab (Misale et al. 2012).

Cancer cells generally exhibit increased heterogeneity and genomic complexity following exposure to treatment, leading to increased intratumor heterogeneity in recurrent tumors (Dagogo-Jack and Shaw 2018). For example, analysis of genomic DNA from recurrent human malignant gliomas reveals a large number of somatic mutations following alkylating agent treatment (Hunter et al. 2006). Higher levels of intratumor heterogeneity have been linked to poorer responses to anticancer treatments. In the case of chronic myeloid leukemia, imatinib, a common treatment, can be rendered ineffective by a variety of distinct point mutations that confer drug resistance. Second-line agents, such as dasatinib and nilotinib, are effective against some mutations, but partial resistance can still occur (Gambacorti-Passerini et al. 2011; Burgess et al. 2006). Consequently, the responses of recurrent tumors exhibiting increased intratumor heterogeneity to subsequent lines of therapy are often less robust than those observed following the initial treatment of the primary tumor. Therefore, the level of clonal diversity in recurrent tumors, an important indicator of intratumor heterogeneity, is crucial for clinical decision-making and treatment efficacy. A comprehensive understanding of this clonal diversity has significant implications for guiding treatment strategies².

To investigate clonal diversity, we utilize a stylized two-type birth-death process model. In this model, we examine a population of drug-sensitive cancer cells that undergo continuous reduction in population size during therapy. These drug-sensitive cells accumulate driver mutations at a small, individual constant rate. Cells with driver mutations become resistant to the therapy, enabling them to (potentially) escape extinction and lead to cancer recurrence. This work builds upon previous research (Foo and Leder 2013; Foo et al. 2014; Hanagal et al. 2022). Foo and Leder (2013) examine the pathway by which cancer cells escape treatment. They obtain a uniform in time approximation for the sample paths as the initial size of the tumor approaches infinity. Additionally, they study two important times in the course of cancer recurrence: (1) the time at which the total population size (drug-sensitive cells plus drug-resistant cells) begins to rebound, and (2) the first time at which the size of drug-resistant cells exceeds that of drug-sensitive cells (crossover time). In Foo et al. (2014), the authors extend the result of Foo and Leder (2013) by including random mutational fitness advantage. They consider a more general setting in which each mutation results in a mutant with a birth rate sampled randomly from a distribution. The authors obtain a functional central limit result for the mutant cell process and then establish a central limit theorem for the crossover time. In Hanagal et al. (2022), the authors investigate

¹ Another main cause of drug resistance is intrinsic resistance, which refers to the presence of resistant cells that exist prior to the treatment. In this work, we focus on the modeling of acquired resistance.

² While this research is motivated by cancer recurrence, the findings can be applied to other areas such as pest control, parasitic infection treatment, and treatment of other diseases caused by viruses and bacteria.

the large deviations of cancer recurrence timing. The authors obtain a convergence in probability result for the recurrence time and then apply a large deviations analysis to the event of early recurrence (a similar result is obtained for the crossover time). The authors also obtain the most likely number of mutant clones at cancer recurrence through optimizing the large deviations rate.

Our work is also related to a body of literature that explores the path to extinction of a biological group and its escape from extinction. Iwasa et al. (2003) use a multi-type branching process model to study the escape dynamics of a biological group from biomedical intervention. The authors obtain the probability of a successful escape under different scenarios, considering factors such as the number of point mutations required to confer resistance and the distribution of mutants before intervention. Jagers et al. (2007a) investigate the path to extinction of a subcritical Markov branching process. They obtain a convergence of finite dimensional distributions result for the path when the time is scaled to $[0, 1]$, where 0 denotes the starting time, and 1 denotes the extinction time. They (Jagers et al. 2007b) further extend the result of Jagers et al. (2007a) by considering a more general branching process. Sagitov and Serra (2009) study the escape from the extinction of a Bienayme-Galton-Watson process. They obtain the limit process conditioned on successful escape as the mutation rate goes to zero. In Sehl et al. (2011), the author employs a birth-death process to investigate the extinction times of cancer cells and normal cells in response to a therapy. Their findings can be used to evaluate a therapy's safety and efficacy. Lastly, Avanzini and Antal (2019) use a branching process model to study cancer recurrence resulting from latent metastases.

Another related stream of literature focuses on the intratumor heterogeneity induced by mutations in the tumor cell population. Previous studies (Coldman and Goldie 1986; Michelson et al. 1989; Komarova 2006; Haeno et al. 2009; Durrett and Moseley 2009; Durrett et al. 2011; Keller and Antal 2015; Cheek and Antal 2018) have extensively examined the mutation of cancer cells and their resulting drug resistance. Of particular relevance to our work is the study by Durrett et al. (2011), where the authors examine the intratumor heterogeneity of a tumor during its expansion. They investigate a multi-type branching process model in which each mutation results in a random, additive change in the cell's birth rate. They study both between-generation heterogeneity (where cells with the same number of mutations are grouped together as a generation) and within-generation heterogeneity in the first generation of cells (comprised of cells with only one mutation). For the latter part, they investigate two metrics to measure heterogeneity: (1) Simpson's Index and (2) the fraction of cells that belong to the largest clone.

In this article, we investigate two indices of clonal diversity: the number of mutant clones and the Simpson's Index of mutant clones at cancer recurrence. In Section 3, we obtain the limit of the expectation of the scaled number of mutant clones and the scaled Simpson's Index (at a deterministic time close to the cancer recurrence time) per Lemma 1 and Proposition 1 respectively. In Section 4, we study the number of mutant clones and the Simpson's Index of mutant clones conditioned on early recurrence in the special case of deterministically decaying sensitive population. In Proposition 3, we show that the distribution of the number of mutant clones at cancer recurrence conditioned on early recurrence stochastically dominates that without conditioning

in the large population limit. In Lemma 2, we obtain the limit of the expectation of the scaled number of mutant clones conditioned on early recurrence. We find that the expected number of clones conditioned on early recurrence is higher than that without conditioning. In Proposition 4, we show that, conditioned on early recurrence, the number of clones generated in any given sufficiently small time period is concentrated at a larger number than would be expected without conditioning. In Theorem 2, we obtain the limit of the expectation of the scaled Simpson's Index of mutant clones conditioned on early recurrence. Simulation results indicate that the Simpson's Index conditioned on early recurrence is smaller than that without conditioning. All of our findings suggest that early recurrence is associated with higher clonal diversity. In particular, our results indicate that early recurrence is driven by a larger than expected number of mutations which leads to a more diverse resistant population.

The remainder of this paper is organized as follows. In Section 2, we describe our model and present important results from previous works. In Section 3, we present results on diversity indices without conditioning. In Section 4, we present results on diversity indices conditioned on early recurrence. In Section 5, we summarize our main findings, along with study limitations and potential future directions. In Sections 6 to 8, we present the proofs of our main results.

2 Models and Previous Results

In this section, we describe a two-type birth-death process model, which has been employed in previous studies to examine the progression of cancer under treatment (see, for example, Foo and Leder (2013) and Foo et al. (2014)).

Consider a subcritical birth-death process $(Z_0^n(t))_{t \geq 0}$ with birth rate r_0 , death rate d_0 and net growth rate $\lambda_0 = r_0 - d_0 < 0$. Z_0^n represents the population size of *drug-sensitive cells* under a certain treatment. At time 0, the initial population size is $Z_0^n(0) = n$. We assume that at time t , drug-sensitive cells give birth to a *drug-resistant mutant* and a drug-sensitive cell at rate $Z_0^n(t) \mu n^{-\alpha}$ for $\alpha \in (0, 1)$ ³. Each of these mutations results in the creation of a distinct clone (each mutation is distinct under the infinite sites approximation). We denote by $I_n(t)$ the number of mutant clones generated in the time period $(0, t)$ and $X_{i,n}(t)$ the population size of the i -th clone at time t , where $i \leq I_n(t)$. Note that the mutant clones are ordered at random, not in chronological order by when the mutation occurred. Let t_i represent the time at which the i -th clone is generated. Then, we assume $(X_{i,n}(s))_{s \geq 0} = (1_{s \geq t_i} Z^{(i)}((s - t_i) \vee 0))_{s \geq 0}$, where $(Z^{(i)})_{i \geq 1}$ are i.i.d. Yule processes⁴ with a birth rate of $r_1 = \lambda_1 > 0$ and a death rate of $d_1 = 0$. We denote the population size of all (mutant) clones at time t by $(Z_1^n(t))_{t \geq 0}$, where $Z_1^n(t) = \sum_i X_{i,n}(t)$. Then Z_1^n is a supercritical branching process

³ We scale the mutation rate with n such that the individual mutation rate approaches zero as the initial population size $n \rightarrow \infty$. This corresponds to the large-population, small-mutation regime.

⁴ Neglecting resistant cell death is a simplification of the biological reality. We make this choice primarily to maintain analytical tractability. Should cell death be incorporated, our analytical results would apply to the skeleton subpopulation of resistant cells whose descendants do not go extinct. Moreover, in Section 4.1, we use simulations to demonstrate that our main findings are robust to the relaxation of this assumption.

with immigration. For each $n \geq 1$, the processes (Z_0^n, Z_1^n) are defined on a common probability space $(\Omega_n, \mathcal{F}^n, \mathbb{P}_n)$. For ease of notation we will write $\mathbb{P}$ instead of $\mathbb{P}_n$.

Define $z_1^n(t) = \mathbb{E}Z_1^n(t)$, then (cf. Athreya and Ney [n.d.](#)):

$$z_1^n(t) = \frac{\mu}{\lambda_1 - \lambda_0} n^{1-\alpha} e^{\lambda_1 t} \left(1 - e^{(\lambda_0 - \lambda_1)t}\right).$$

We also present a few useful results for a birth-death process starting from a single cell. Let $Z = \{Z(t), t \geq 0\}$ denote a birth-death process where $Z(0) = 1$ and each individual cell has birth rate r , death rate d , and net growth rate $\lambda = r - d$. The moment generating function of $Z(t)$ is given by

$$\phi_t(\theta) = \mathbb{E} \exp(\theta Z(t)) = \begin{cases} \frac{d(e^\theta - 1) - e^{-\lambda t}(re^\theta - d)}{r(e^\theta - 1) - e^{-\lambda t}(re^\theta - d)}, & \theta < \bar{\theta}_t \\ \infty & \theta \geq \bar{\theta}_t \end{cases} \quad (1)$$

where

$$\bar{\theta}_t \doteq \log \left(\frac{re^{\lambda t} - d}{re^{\lambda t} - r} \right) \quad (2)$$

(see page 109 of Athreya and Ney [n.d.](#)). Throughout this paper, we will repeatedly use ϕ_t to denote the moment generating function of $Z(t)$. If we let $\Psi_1 = \frac{de^{\lambda t} - d}{re^{\lambda t} - d}$ and $\Psi_2 = \frac{re^{\lambda t} - r}{re^{\lambda t} - d}$, then (see page 6 of Durrett [2014](#))

$$\mathbb{P}(Z(t) = 0) = \Psi_1, \quad \mathbb{P}(Z(t) = n) = (1 - \Psi_1)(1 - \Psi_2)\Psi_2^{n-1} \quad \text{for } n \geq 1.$$

We define the recurrence time as

$$\gamma_n(a) = \inf\{t \geq 0 : Z_1^n(t) > an\}$$

for $a > 0$. The recurrence time represents the first time that the mutant cell population exceeds a proportion a of the initial population size of drug-sensitive cells. We will often be interested in $\gamma_n(1)$, and use the notation $\gamma_n \equiv \gamma_n(1)$. We denote by $\zeta_n(a)$ the deterministic approximation to $\gamma_n(a)$, which is the unique value of t such that $z_1^n(t) = an$. It has been established in Hanagal et al. ([2022](#)) that

$$\zeta_n(a) - \frac{1}{\lambda_1} \log \left(\frac{an^\alpha (\lambda_1 - \lambda_0)}{\mu} \right) \rightarrow 0 \quad (3)$$

as $n \rightarrow \infty$, and $\gamma_n(a) - \zeta_n(a) \rightarrow 0$ in probability. Note that we will use the notation $\zeta_n \equiv \zeta_n(1)$. Consider the event of early recurrence such that recurrence happens y units of time earlier than the deterministic limit, i.e., $\{\gamma_n(a) \leq \zeta_n(a) - y\}$. We have the following large deviations result from Hanagal et al. ([2022](#)).

Theorem 1 Assume that $\alpha \in (0, 1)$, then for $y > 0$,

$$\lim_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P}(\gamma_n(a) \leq \zeta_n(a) - y) = - \sup_{\theta \in (0,1)} \left[\frac{\mu \theta e^{y\lambda_1}}{\lambda_1 - \lambda_0} - \mu \theta \int_0^\infty \frac{e^{\lambda_0 s}}{e^{\lambda_1 s} - \theta} ds \right].$$

Note that the supremum on the right-hand side of the equation in Theorem 1 is positive, and the solution to this optimization problem will be used frequently throughout the rest of the paper.

Definition 2.1 We define θ_y^* to be the unique solution to the following equation

$$\frac{e^{\lambda_1 y}}{\lambda_1 - \lambda_0} = \int_0^\infty \frac{e^{\lambda_1 s}}{(e^{\lambda_1 s} - \theta)^2} e^{\lambda_0 s} ds. \quad (4)$$

Note that θ_y^* satisfies the first order optimality condition of the optimization problem that appears in Theorem 1. Moreover, θ_y^* is a positive number because when $\theta = 0$, the left-hand side of (4) is larger than the right-hand side of (4), and the right-hand side of (4) is an increasing function of θ which goes to infinity as $\theta \rightarrow 1$.

Throughout this work we will use the following notation for the asymptotic behavior of positive functions:

$$\begin{aligned} f(t) &\sim g(t) && \text{if } f(t)/g(t) \rightarrow 1 \text{ as } t \rightarrow \infty, \\ f(t) &= o(g(t)) && \text{if } f(t)/g(t) \rightarrow 0 \text{ as } t \rightarrow \infty, \\ f(t) &= O(g(t)) && \text{if } f(t) \leq Cg(t) \text{ for all } t, \\ f(t) &= \Theta(g(t)) && \text{if } cg(t) \leq f(t) \leq Cg(t) \text{ for all } t, \end{aligned}$$

where C and c are positive constants.

3 Results on Diversity Indices Without Conditioning

In this section, we examine the number of mutant clones and the Simpson's Index of mutant clones at the deterministic time $\zeta_n(1)$ which is a very good approximation of the cancer recurrence time $\gamma_n(1)$. Recall that we denote by $I_n(t)$ the number of mutant clones generated in the time period $(0, t)$. We first obtain the limit of the expected scaled number of mutant clones.

Lemma 1

$$\lim_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \mathbb{E}[I_n(\zeta_n)] = -\frac{\mu}{\lambda_0}.$$

This result tells us that the number of mutant clones at the deterministic limit of recurrence time is of order $\Theta(n^{1-\alpha})$. The limit increases in μ and decreases in $|\lambda_0|$, which is expected as a higher mutation rate or a lower decaying rate of sensitive cells leads to more mutant clones.

We then investigate the Simpson's Index of mutant clones. Simpson's Index is a measure of diversity that considers both the number of clones and their relative sizes. As the number of clones and the evenness of their sizes increase, so does the diversity. Mathematically, Simpson's Index represents the probability that two randomly chosen cells from the mutant cell population belong to the same clone. Therefore, the Simpson's Index ranges between 0 and 1. The Simpson's Index is close to 1 (indicating low diversity) if a few mutant clones dominate the mutant population. If there are many clones, all of similar sizes, the Simpson's Index is near zero (indicating high diversity).

Recall that $X_{i,n}(\zeta_n)$ denotes the population size of the i -th mutant clone at time ζ_n , and the mutant clones are ordered at random, not in chronological order by when the mutation occurred. For simplicity, we will omit the notation (ζ_n) . Then the Simpson's index is computed by

$$R_n(\zeta_n) = \sum_{i=1}^{I_n(\zeta_n)} \left(\frac{X_{i,n}}{Z_1^n(\zeta_n)} \right)^2, \quad (5)$$

and we define $R_n(\zeta_n) = 0$ when $Z_1^n(\zeta_n) = 0$. We have the following proposition for the large n behavior of the expected Simpson's Index.

Proposition 1

$$\lim_{n \rightarrow \infty} n^{1-\alpha} \mathbb{E}[R_n(\zeta_n)] = \frac{2(\lambda_1 - \lambda_0)^2}{\mu(2\lambda_1 - \lambda_0)}.$$

This result indicates that the expected Simpson's Index of mutant clones at the deterministic limit of recurrence time is of the order $\Theta(n^{\alpha-1})$. Recall that the expected number of mutant clones is of the order $\Theta(n^{1-\alpha})$. If all clones are of equal size, achieving maximum diversity as per the AM-QM inequality, the corresponding Simpson's Index also follows the order $\Theta(n^{\alpha-1})$. This aligns with the expected Simpson's Index of mutant clones at time ζ_n , suggesting a high evenness of clone sizes at cancer recurrence.

Examining the limit of the scaled Simpson's Index, we notice that the limit decreases in μ , as a higher mutation rate results in a greater number of mutant clones, which leads to a lower Simpson's Index. We also notice that the limit increases in λ_1 , which is owing to the fact that a higher growth rate of mutants results in larger clone sizes and a lesser number of clones at cancer recurrence, resulting in a higher Simpson's Index.

Remark 3.1 The results presented in this section can be used to construct a set of estimators for the model parameters using the method of moments. Readers interested in further details may refer to another paper by the authors, which demonstrates that the model parameters can be estimated from single patient, single time-point sequencing data of recurrent tumors (Leder et al. 2024).

4 Results on Diversity Indices Conditioned on Early Recurrence

In this section, we examine the number of mutant clones and the Simpson's Index of mutant clones at cancer recurrence conditioned on the event of early recurrence ($\{\gamma_n \leq \zeta_n - y\}$). Note that throughout this section y is a positive number independent of n . We compare results conditioned on early recurrence to those without conditioning. Our goal is to determine whether early recurrence is primarily caused by a single mutation, leading to an unusually large family of drug-resistant cells (resulting in low clonal diversity), or if it is instead attributed to an abnormally high number of mutations causing drug resistance (resulting in high clonal diversity).

To derive analytical results conditioned on early recurrence, we impose a strong simplifying assumption that the population of sensitive cells follows a deterministic exponential decay (i.e., $Z_0^n(t) = z_0^n(t)$, where, abusing the notation, $z_0^n(t) = \mathbb{E}[Z_0^n(t)] = ne^{\lambda_0 t}$ in the original model). Although this neglects stochastic fluctuations in the sensitive cell subpopulation, it captures the first-order system dynamics, as the mean trajectory provides a uniform-in-time approximation for the sensitive cells (see Lemma 3 in Foo and Leder (2013)). We also note that this assumption does not affect results presented in previous sections, including Theorem 1, Lemma 1 and Proposition 1. Therefore, we use the same notation introduced in previous sections. In addition, we have verified through extensive numerical simulation that our main results remain valid in more general models where the sensitive cell population follows a birth-death process and resistant cells have a non-zero death rate.

4.1 Number of Mutant Clones Conditioned on Early Recurrence

Because of the assumption that sensitive cells have deterministic exponential decay, we can study the distribution of the number of clones at cancer recurrence. For non-negative integer valued random variables X and Y , their total variation distance is given by

$$TV(X, Y) = \sum_{k=0}^{\infty} |P(X = k) - P(Y = k)|.$$

We first show that under the assumption $\alpha \in \left(\frac{\lambda_1}{\lambda_1 - \lambda_0}, 1\right)$, $I_n(\gamma_n)$ is close to $I_n(\zeta_n)$ in total variation distance.

Proposition 2 *If $\alpha \in \left(\frac{\lambda_1}{\lambda_1 - \lambda_0}, 1\right)$, then*

$$\lim_{n \rightarrow \infty} TV(I_n(\gamma_n), I_n(\zeta_n)) = 0.$$

The assumption on α is biologically plausible, as mutation rates in driver genes are typically low, often resulting in values of α close to 1. Mathematically, if the assumption is not satisfied, i.e., $\alpha \leq \frac{\lambda_1}{\lambda_1 - \lambda_0}$, then it implies a non-negligible probability that a mutation occurs within an $O(1)$ time window around the time ζ_n . In such cases, the result in Proposition 2 would no longer hold.

Because sensitive cells have deterministic exponential decay, we know that $(I_n(t))_{t \geq 0}$ is a non-homogeneous Poisson process, which gives us that

$$\mathbb{P}(I_n(\zeta_n) = k) = \frac{\lambda_n^k e^{-\lambda_n}}{k!},$$

where $\lambda_n = -\frac{\mu n^{1-\alpha}}{\lambda_0} (1 - e^{\lambda_0 \zeta_n})$. Proposition 2 tells us that the distribution of the number of mutant clones at cancer recurrence is close to a Poisson distribution with mean λ_n .

We then show that the number of clones at the cancer recurrence time conditioned on the event of early recurrence $(\{\gamma_n \leq \zeta_n - y\})$ stochastically dominates that without conditioning asymptotically under assumption on α .

Proposition 3 *If $\alpha \in \left(\frac{\lambda_1}{\lambda_1 - \lambda_0} \vee \frac{1}{2}, 1\right)$, then*

$$\liminf_{n \rightarrow \infty} \inf_{x > 0} (\mathbb{P}(I_n(\gamma_n) \geq x | \gamma_n < \zeta_n - y) - \mathbb{P}(I_n(\gamma_n) \geq x)) \geq 0.$$

This finding sheds light on the effect of early recurrence on the distribution of mutant clones at cancer recurrence. If a patient experiences an early cancer recurrence, the number of mutant clones in the recurring tumor is very likely to be higher than expected. As a result, the recurrent tumor is more likely to be resistant to second-line medications, which should be taken into account when determining future treatment options.

We conjecture that Proposition 3 holds under more general settings in which the sensitive cell population evolves as a birth-death process and resistant cells have a non-zero death rate. To support this conjecture, we perform extensive numerical simulations. Specifically, We simulate the system using the Gillespie algorithm, which generates exact realizations of the sequence of events (e.g., birth, death, mutation) and their precise occurring times. Representative output from these simulations is shown in Figure 1.

From Proposition 3, we know that the number of mutant clones at cancer recurrence conditioned on early recurrence stochastically dominates that without conditioning in the large population limit. Because a stochastically larger random variable has a larger expectation, the number of mutant clones at cancer recurrence conditioned on early recurrence should have a higher expectation than that without conditioning. In Lemma 2, we obtain the expectation of the scaled number of mutant clones at $\zeta_n - y$ conditioned on early recurrence. For simplicity, we denote by $A_{n,y}$ the event of early recurrence, i.e.,

$$A_{n,y} = \{\omega : \gamma_n < \zeta_n - y\},$$

and the corresponding conditional probability measure by $\mathbb{P}_{A_{n,y}}(\cdot) = \mathbb{P}(\cdot | A_{n,y})$.

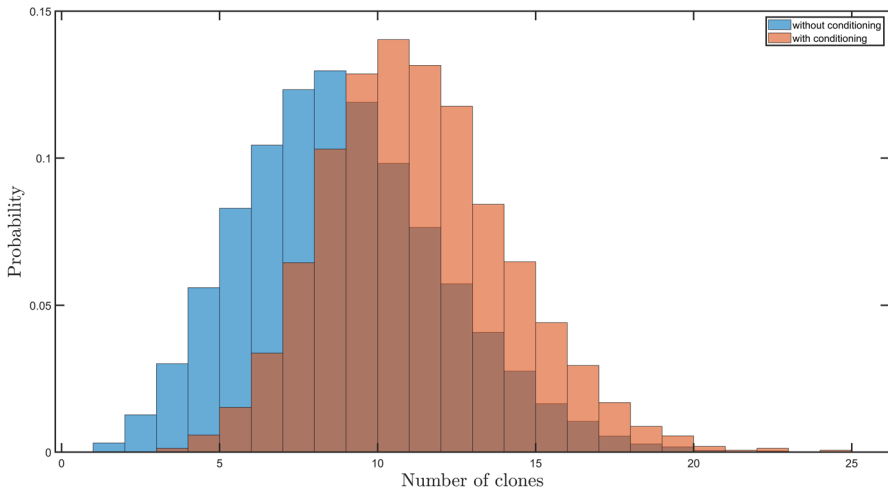


Fig. 1 Scaled histogram of number of clones present at recurrence for simulations conditioned on early recurrence (light brown) and unconditioned simulations (blue). Histogram is based on 10^4 simulations and all simulations used the model parameter set $r_0 = 1$, $d_0 = 1.2$, $r_1 = 1$, $d_1 = 0.8$, $\mu = 0.5$, $\alpha = 0.6$, $y = 1$, $n = 1,000$ and $a = 1$

Lemma 2

$$\lim_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \mathbb{E}_{A_{n,y}} [I_n(\zeta_n - y)] = \mu \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta_y^*} e^{\lambda_0 s} ds,$$

where θ_y^* is defined in (4).

Observe that θ_y^* increases in y and thus $\mu \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta_y^*} e^{\lambda_0 s} ds$ increases in y , which implies that for any $y > 0$,

$$\mu \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta_y^*} e^{\lambda_0 s} ds > -\frac{\mu}{\lambda_0}.$$

This result is consistent with Proposition 3. Moreover, we can see that the recurrence time is an indicator of the clonal diversity at cancer recurrence.

Lastly, we investigate the number of clones that are generated in a given time period over the course of the treatment. Let $I_{n,(t_1,t_2)}$ denote the number of clones generated in the time period (t_1, t_2) . Let $B_{n,y,(t_1,t_2)}$ denote the number of mutants at time $\zeta_n - y$ which belong to those clones generated in the time period (t_1, t_2) . For simplicity, we let

$$\bar{I}_{n,(t_1,t_2)} = \mathbb{E} [I_{n,(t_1,t_2)}] = \mu n^{1-\alpha} \int_{t_1}^{t_2} e^{\lambda_0 t} dt, \quad (6)$$

and

$$\bar{B}_{n,y,(t_1,t_2)} = \mathbb{E} [B_{n,y,(t_1,t_2)}].$$

Recall that we denote by $A_{n,y}$ the event of early recurrence $\{\gamma_n < \zeta_n - y\}$, and the corresponding conditional probability measure by $\mathbb{P}_{A_{n,y}}(\cdot) = \mathbb{P}(\cdot | A_{n,y})$.

We are interested in the number $I_{n,(t_1,t_2)}$ conditioned on early recurrence. We first show the following lemma.

Lemma 3 For any $\epsilon > 0$,

$$\limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P}_{A_{n,y}} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \notin (1 + \delta^* - \epsilon, 1 + \delta^* + \epsilon) \right) < 0,$$

where δ^* is given by

$$\delta^* = \frac{\int_{t_1}^{t_2} \frac{e^{\lambda_1 s}}{(e^{\lambda_1 s} - \theta_y^*)^2} e^{\lambda_0 s} ds}{\int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds} - 1 > 0,$$

and θ_y^* is defined in (4).

This result tells us that the number of mutants at time $\zeta_n - y$ which belong to those clones generated in the time period (t_1, t_2) is concentrated around $(1 + \delta^*) \bar{B}_{n,y,(t_1,t_2)}$.

We then analyze the number $I_{n,(t_1,t_2)}$ conditioned on the event

$$A_{n,y,\epsilon}^{t_1,t_2} = \{B_{n,y,(t_1,t_2)} \in ((1 + \delta^* - \epsilon) \bar{B}_{n,y,(t_1,t_2)}, (1 + \delta^* + \epsilon) \bar{B}_{n,y,(t_1,t_2)})\}.$$

We have the following lemma.

Lemma 4 Assume that $t_2 - t_1 < -\frac{1}{\lambda_1} \log \left(\frac{1}{2 - \theta_y^*} \right)$. For any $\sigma > 0$, there exists $\bar{\epsilon} > 0$ such that when $0 < \epsilon < \bar{\epsilon}$,

$$\limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P}_{A_{n,y,\epsilon}^{t_1,t_2}} \left(\frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \notin (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma) \right) < 0,$$

where

$$\kappa^* = \frac{\int_{t_1}^{t_2} \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta_y^*} e^{\lambda_0 s} ds}{\int_{t_1}^{t_2} e^{\lambda_0 s} ds} - 1.$$

This result tells us that conditioned on the event that the number of mutants is concentrated around $(1 + \delta^*) \bar{B}_{n,y,(t_1,t_2)}$, the number of clones generated in the time period (t_1, t_2) is concentrated around $(1 + \kappa^*) \bar{I}_{n,(t_1,t_2)}$.

From Lemma 3 and 4, we can obtain the desired result.

Proposition 4 Assume that $t_2 - t_1 < -\frac{1}{\lambda_1} \log \left(\frac{1}{2-\theta_y^*} \right)$. For any $\sigma > 0$,

$$\limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P}_{A_{n,y}} \left(\frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \notin (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma) \right) < 0,$$

where κ^* is defined in Lemma 4.

Note that δ^* and κ^* depend on t_1 and t_2 , and we omit the dependence in their notations for simplicity. Fix t_1 , we can obtain that

$$\lim_{t_2 \downarrow t_1} 1 + \kappa^* = \frac{e^{\lambda_1 t_1}}{e^{\lambda_1 t_1} - \theta_y^*}, \text{ and} \quad (7)$$

$$\lim_{t_2 \downarrow t_1} 1 + \delta^* = \left(\frac{e^{\lambda_1 t_1}}{e^{\lambda_1 t_1} - \theta_y^*} \right)^2. \quad (8)$$

Hence, conditioned on early recurrence, Lemma 3 tells us that the number of mutants at time $\zeta_n - y$ which belong to those clones generated in an infinitesimal time period $(t_1, t_1 + dt)$ is larger than that without conditioning by the approximate factor $\left(\frac{e^{\lambda_1 t_1}}{e^{\lambda_1 t_1} - \theta_y^*} \right)^2$. Moreover, from (7) and (8), we can specify the contribution to such an increase in the number of mutants from (1) the increase in the number of clones, and (2) the increase in the average clone size. Our results (Lemma 3 and Proposition 4) indicate that both factors contribute equally to the increase in the number of mutants by the approximate factor $\frac{e^{\lambda_1 t_1}}{e^{\lambda_1 t_1} - \theta_y^*}$.

4.2 Simpson's Index Conditioned on Early Recurrence

In this section, we investigate the Simpson's Index of mutant clones at time $\zeta_n - y$ conditioned on early recurrence. Recall that we denote by $A_{n,y}$ the event of early recurrence $\{\gamma_n < \zeta_n - y\}$, and the corresponding conditional probability measure by $\mathbb{P}_{A_{n,y}}(\cdot) = \mathbb{P}(\cdot | A_{n,y})$. Let $R_{n,y}$ be the Simpson's Index of mutant clones at time $\zeta_n - y$. We have the following theorem.

Theorem 2

$$\lim_{n \rightarrow \infty} n^{1-\alpha} \mathbb{E}_{A_{n,y}} [R_{n,y}] = \frac{2(\lambda_1 - \lambda_0)^2}{\mu} e^{-2\lambda_1 y} \int_0^\infty \frac{e^{-(2\lambda_1 - \lambda_0)s}}{(1 - \theta_y^* e^{-\lambda_1 s})^3} ds \doteq S_c(y), \quad (9)$$

where θ_y^* is defined in (4).

This result tells us that the Simpson's Index of mutant clones at $\zeta_n - y$ conditioned on early recurrence is of order $\Theta(n^{\alpha-1})$ as well. The limit is associated with the value

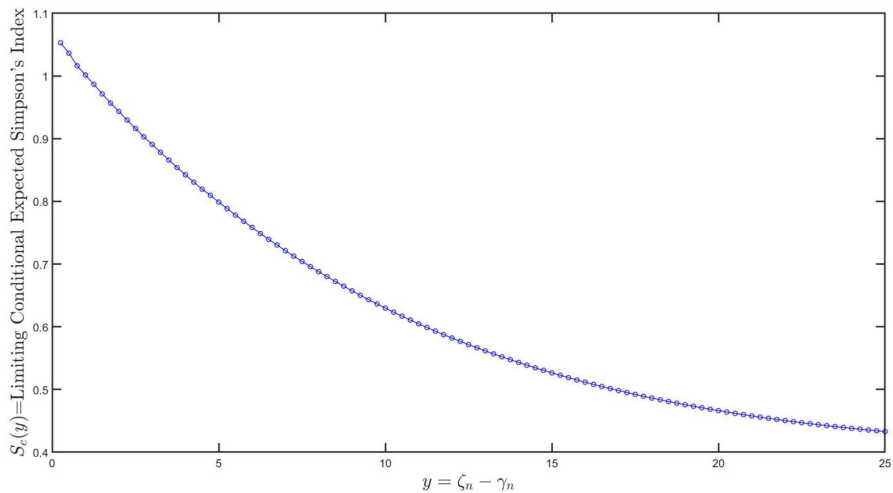


Fig. 2 Plot of $S_c(y)$ versus y , $S_c(y)$ is defined in (9). Model Parameters: $r_0 = 1$, $d_0 = 1.2$, $\lambda_0 = -0.2$, $\lambda_1 = 0.2$, $\mu = 0.5$, $\alpha = 0.6$

Table 1 Simpson's Index at Early Recurrence. Values represent the average Simpson's Index of mutant clones computed from 1,000 simulations for each given value of $\zeta_n - y$. Model parameters: $r_0 = 1$, $d_0 = 1.2$, $r_1 = 1.2$, $d_1 = 1$, $\mu = 1$, $\alpha = 0.6$, $n = 100,000$ and $a = 1$

$\zeta_n - y$	29.9	29.8	29.7	29.6	29.5
Simpson's Index	0.0302	0.0301	0.0298	0.0294	0.0291

of y . We conjecture that the limit $S_c(y)$ decreases in y , and it is always smaller than that obtained in Proposition 1. However, we were not able to prove this result. Our conjecture is supported by numerical results, an example of which is given in Figure 2. We further conjecture that the expected Simpson's Index of mutant clones conditioned on early recurrence decreases in the value of y under more general settings (sensitive cell population evolves as a birth-death process and resistant cells have a non-zero death rate). Numerical simulations provide support for this conjecture, as shown in Table 1.

Our analysis of both clonal diversity indices reveals that early recurrent tumors are more likely to exhibit higher clonal diversity. This suggests that early recurrence is primarily driven by a larger number of mutations, rather than the successful expansion of a single mutation. Furthermore, our findings highlight that the time of recurrence serves as an important indicator of clonal diversity at cancer recurrence.

5 Concluding Comments

In this work, we have examined the clonal diversity of mutant clones at cancer recurrence. We obtained the expectation of the number of mutant clones and the Simpson's Index at cancer recurrence with and without conditioning on early recurrence. We

Table 2 The Number of Clones and Simpson's Index at Early Recurrence (random mutational fitness effects). Values represent the average number of clones and Simpson's Index computed from 100 simulations for each given value of $\zeta_n - y$. Model parameters: $r_0 = 1$, $d_0 = 1.2$, $r_1 \sim \text{Uniform}[0, 0.2]$, $d_1 = 0$, $\mu = 0.5$, $\alpha = 0.6$, $n = 100,000$ and $a = 1$

$\zeta_n - y$	43	42.5	42	41.5
Number of Clones	257.9800	255.4400	261.1100	262.5300
Simpson's Index	0.0426	0.0443	0.0444	0.0440

Table 3 The Number of Clones and Simpson's Index at Early Recurrence ($\alpha = 1$). Values represent the average number of clones and Simpson's Index computed from 1,000 simulations for each given value of $\zeta_n - y$. Model parameters: $r_0 = 1$, $d_0 = 1.2$, $r_1 = 0.2$, $d_1 = 0$, $\mu = 0.5$, $\alpha = 1$, $n = 100,000$ and $a = 1$

$\zeta_n - y$	57	56	55	54
Number of Clones	3.5650	3.6170	3.8020	3.8990
Simpson's Index	0.5504	0.5479	0.5258	0.5201

showed that the number of mutant clones at cancer recurrence conditioned on early recurrence stochastically dominates that without conditioning in the large population limit. In addition we have derived an expression for the large population limit of Simpson's Index conditioned on early recurrence. Our findings suggest that an earlier recurrent tumor is more likely to have a higher clonal diversity. Furthermore, our results suggest that early recurrence is most likely to be generated by a larger number of mutations, as opposed to the successful growth of a single mutation.

Our work has several important limitations. Here, we highlight three of the most significant. First, our main findings rely on the assumption that resistant cells have equal fitness. In reality, distinct mutant clones may have different fitnesses (Foo et al. 2014), which complicates the evolutionary dynamics of recurrence. Introducing fitness variation adds a potential mechanism where early, high-fitness mutations drive rapid expansion and early recurrence. This mechanism would likely reduce overall clonal diversity, a pattern we observe via a lower Simpson's Index in simulations (see Table 2). However, we conjecture that our main asymptotic results remain valid, as prior analyses suggest that intratumor heterogeneity depends primarily on the maximum achievable fitness (Durrett et al. 2011). A more nuanced and biologically relevant question is whether, for plausible tumor sizes, early recurrence correlates with higher or lower clonal diversity.

Second, our model assumes the per-sensitive-cell mutation rate scales as $\mu n^{-\alpha}$ with $\alpha < 1$. This scaling ensures that, in the limit, recurrence occurs with probability one. Consequently, our analysis does not encompass cancer types with a low probability of recurrence, which corresponds to the critical scaling $\alpha = 1$. Simulations in this alternate regime (Table 3) reveal that, conditional on recurrence occurring, the number of resistant clones is significantly smaller and the Simpson's Index is significantly larger than in the $\alpha < 1$ case. Nevertheless, we observe similar qualitative trends in diversity: conditioning on early recurrence leads to an increase in the expected number of clones and a decrease in the expected Simpson's Index.

Third, our model does not incorporate physical or nutritional constraints that would impose a carrying capacity on tumor size (Leder et al. 2025). In reality, such capacity constraints could reduce cellular growth rates as the tumor burden increases. However, we conjecture that our main findings remain relevant. Most resistant mutations are likely to arise early during therapy when the sensitive cell population is large and growth is restricted. By the time therapy significantly reduces the tumor burden, potentially relaxing density constraints and allowing for faster mutant expansion, a substantial number of resistant clones have already been founded. This initial diversity makes it difficult for any single clone to achieve dominance, thereby preserving the heterogeneity described by our model.

We believe these limitations highlight several promising directions for future research, such as incorporating fitness variation and alternative mutation-rate scalings, which we leave for future work.

6 Proofs of Results in Section 3

6.1 Proof of Lemma 1

Proof This result can be derived from direct calculation and thus the proof is omitted. $\square$

6.2 Proof of Proposition 1

Proof Note that in this proof for ease of notation, we will use the notation $R_n \equiv R_n(\zeta_n)$. Recall that $I_n(\zeta_n)$ represents the number of clones generated in the time period $(0, \zeta_n)$. We define

$$\tilde{R}_n = \sum_{i=1}^{I_n(\zeta_n)} \left(\frac{X_{i,n}}{n} \right)^2. \quad (10)$$

We can compute that

$$\begin{aligned} n^{1-\alpha} \mathbb{E} [\tilde{R}_n] &= n^{1-\alpha} \mathbb{E} \left[\sum_{i=1}^{I_n(\zeta_n)} \left(\frac{X_{i,n}}{n} \right)^2 \right] \\ &= n^{1-\alpha} \mathbb{E} \left[\mathbb{E} \left[\mathbb{E} \left[\sum_{i=1}^{I_n(\zeta_n)} X_{i,n}^2 \middle| I_n(\zeta_n) \right] \middle| Z_0^n \right] \right] \\ &\stackrel{(a)}{=} n^{1-\alpha} \mathbb{E} \left[\int_0^{\zeta_n} \mu n^{-\alpha} Z_0^n(s) \mathbb{E} [(Z(\zeta_n - s))^2] ds \right] \\ &= n^{1-\alpha} \int_0^{\zeta_n} \mu n^{1-\alpha} e^{\lambda_0 s} \mathbb{E} [(Z(\zeta_n - s))^2] ds \end{aligned}$$

$$\begin{aligned}
&\stackrel{(b)}{=} n^{-1-\alpha} \int_0^{\zeta_n} \mu n^{1-\alpha} e^{\lambda_0 s} \left(2e^{2\lambda_1(\zeta_n-s)} - e^{\lambda_1(\zeta_n-s)} \right) ds \\
&\stackrel{(c)}{\rightarrow} \frac{2(\lambda_1 - \lambda_0)^2}{\mu(2\lambda_1 - \lambda_0)} \text{ as } n \rightarrow \infty,
\end{aligned} \tag{11}$$

where we use the uniformity of arrival times for a Poisson process in step (a), $\mathbb{E}[(Z(t))^2] = 2e^{2\lambda_1 t} - e^{\lambda_1 t}$ in step (b), and $e^{\lambda_1 \zeta_n} \sim \frac{\lambda_1 - \lambda_0}{\mu} n^\alpha$ in step (c). Also note that the process $Z(\cdot)$ in step (a) is defined in page 4 before (1) with $d_1 = 0$. We then compute the difference between $\mathbb{E}[\tilde{R}_n]$ and $\mathbb{E}[R_n]$. Recall that we define $R_n = 0$ when $Z_1^n(\zeta_n) = 0$. It allows us to work on the event that $Z_1^n(\zeta_n) > 0$. Hence, for the rest of this section, we condition on the event $\rho_n = \{Z_1^n(\zeta_n) > 0\}$, and denote by $\mathbb{E}_{\rho_n}$ the conditional expectation. We obtain that

$$\begin{aligned}
&n^{1-\alpha} \mathbb{E}_{\rho_n} [\tilde{R}_n - R_n] \\
&= n^{1-\alpha} \mathbb{E}_{\rho_n} \left[\sum_{i=1}^{I_n(\zeta_n)} \left(\frac{X_{i,n}}{Z_1^n(\zeta_n)} \right)^2 - \sum_{i=1}^{I_n(\zeta_n)} \left(\frac{X_{i,n}}{n} \right)^2 \right] \\
&= n^{1-\alpha} \mathbb{E}_{\rho_n} \left[\left| \frac{\sum_{i=1}^{I_n(\zeta_n)} (X_{i,n})^2}{n^2} \left(\frac{n^2}{Z_1^n(\zeta_n)^2} - 1 \right) \right| \right] \\
&\leq n^{1-\alpha} \mathbb{E}_{\rho_n} \left[\left(\sum_{i=1}^{I_n(\zeta_n)} (X_{i,n})^2 \right)^2 / n^4 \right]^{1/2} \mathbb{E}_{\rho_n} \left[\left(\frac{n^2}{Z_1^n(\zeta_n)^2} - 1 \right)^2 \right]^{1/2}.
\end{aligned}$$

We can show that (for some positive constants c_1 and c_2)

$$\begin{aligned}
&\mathbb{E} \left[\left(\sum_{i=1}^{I_n(\zeta_n)} (X_{i,n})^2 \right)^2 / n^4 \right] \\
&= \frac{1}{n^4} \int_0^{\zeta_n} \mu n^{1-\alpha} e^{\lambda_0 s} \mathbb{E}[(Z(\zeta_n - s))^4] ds \\
&\quad + \frac{1}{n^4} \mathbb{E} \left[\frac{\mathbb{E}[I_n(\zeta_n)^2 - I_n(\zeta_n) | Z_0^n]}{\mathbb{E}[I_n(\zeta_n) | Z_0^n]^2} \left(\int_0^{\zeta_n} \mu n^{1-\alpha} Z_0^n(s) \mathbb{E}[(Z(\zeta_n - s))^2] ds \right)^2 \right] \\
&= \frac{1}{n^4} \int_0^{\zeta_n} \mu n^{1-\alpha} e^{\lambda_0 s} \mathbb{E}[(Z(\zeta_n - s))^4] ds \\
&\quad + \frac{1}{n^4} \mathbb{E} \left[\left(\int_0^{\zeta_n} \mu n^{-\alpha} Z_0^n(s) \mathbb{E}[(Z(\zeta_n - s))^2] ds \right)^2 \right] \\
&\sim c_1 n^{-2+2\alpha},
\end{aligned}$$

where we use the fact that $\frac{\mathbb{E}[I_n(\zeta_n)^2 - I_n(\zeta_n)|Z_0^n]}{\mathbb{E}[I_n(\zeta_n)|Z_0^n]^2} = 1$, $\mathbb{E}[(Z(t))^4] \sim 24e^{4\lambda_1 t}$, and

$$\begin{aligned} & \mathbb{E} \left[\left(\int_0^{\zeta_n} Z_0^n(s) \mathbb{E}[(Z(\zeta_n - s))^2] ds \right)^2 \right] \\ &= \mathbb{E} \left[\left(\int_0^{\zeta_n} Z_0^n(s) (2e^{2\lambda_1(\zeta_n - s)} - e^{\lambda_1(\zeta_n - s)}) ds \right)^2 \right] \\ &= \int_0^{\zeta_n} \int_0^{\zeta_n} \mathbb{E} \left[Z_0^n(s) (2e^{2\lambda_1(\zeta_n - s)} - e^{\lambda_1(\zeta_n - s)}) Z_0^n(t) (2e^{2\lambda_1(\zeta_n - t)} - e^{\lambda_1(\zeta_n - t)}) \right] dt ds \\ &= \int_0^{\zeta_n} \int_s^{\zeta_n} \mathbb{E} \left[Z_0^n(s) Z_0^n(s) e^{\lambda_0(t-s)} (2e^{2\lambda_1(\zeta_n - s)} - e^{\lambda_1(\zeta_n - s)}) (2e^{2\lambda_1(\zeta_n - t)} - e^{\lambda_1(\zeta_n - t)}) \right] dt ds \\ &\quad + \int_0^{\zeta_n} \int_0^s \mathbb{E} \left[Z_0^n(t) Z_0^n(t) e^{\lambda_0(s-t)} (2e^{2\lambda_1(\zeta_n - s)} - e^{\lambda_1(\zeta_n - s)}) (2e^{2\lambda_1(\zeta_n - t)} - e^{\lambda_1(\zeta_n - t)}) \right] dt ds \\ &\sim c_2 n^{2+4\alpha}. \end{aligned}$$

We can then obtain that

$$n^{1-\alpha} \mathbb{E}_{\rho_n} \left[\left(\sum_{i=1}^{I_n(\zeta_n)} (X_{i,n})^2 \right)^2 / n^4 \right]^{1/2} \rightarrow \sqrt{c_1}.$$

It remains to analyze

$$\mathbb{E}_{\rho_n} \left[\left(\frac{n^2}{Z_1^n(\zeta_n)^2} - 1 \right)^2 \right] = \mathbb{E}_{\rho_n} \left[\frac{n^4}{Z_1^n(\zeta_n)^4} - \frac{2n^2}{Z_1^n(\zeta_n)^2} + 1 \right].$$

We need to analyze the probability that $Z_1^n(\zeta_n)$ deviates from its mean n . Recall that

$$\phi_t(\theta) = \frac{e^\theta e^{-\lambda_1 t}}{1 - (1 - e^{-\lambda_1 t}) e^\theta}.$$

For $\epsilon \in (0, 1)$, and $\theta > 0$, we have

$$\begin{aligned} \mathbb{P}(Z_1^n(\zeta_n) < (1 - \epsilon)n) &= \mathbb{P}(e^{-\theta Z_1^n(\zeta_n)} > e^{-\theta(1-\epsilon)n}) \\ &\leq \min_{\theta > 0} e^{\theta(1-\epsilon)n} \mathbb{E}[e^{-\theta Z_1^n(\zeta_n)}] \\ &= \min_{\theta > 0} e^{\theta(1-\epsilon)n} \mathbb{E} \left[\exp \left(\frac{\mu}{n^\alpha} \int_0^{\zeta_n} Z_0^n(s) (\phi_{\zeta_n - s}(-\theta) - 1) ds \right) \right]. \end{aligned} \tag{12}$$

The expectation term in (12) can be decomposed into a mean behavior term and a fluctuation term:

$$\begin{aligned} & \mathbb{E} \left[\exp \left(\frac{\mu}{n^\alpha} \int_0^{\zeta_n} Z_0^n(s) (\phi_{\zeta_n-s}(-\theta) - 1) ds \right) \right] \\ &= \exp \left(\frac{\mu}{n^{\alpha-1}} \int_0^{\zeta_n} e^{\lambda_0 s} (\phi_{\zeta_n-s}(-\theta) - 1) ds \right) \\ & \quad \times \mathbb{E} \left[\exp \left(\frac{\mu}{n^\alpha} \int_0^{\zeta_n} (\phi_{\zeta_n-s}(-\theta) - 1) (Z_0^n(s) - n e^{\lambda_0 s}) ds \right) \right]. \end{aligned}$$

By Proposition 1 of Hanagal et al. (2022), we can safely discard the fluctuation term, and focus on the remaining terms in the exponential expression of (12):

$$\begin{aligned} & \theta (1 - \epsilon) n + \frac{\mu}{n^\alpha} \int_0^{\zeta_n} n e^{\lambda_0 s} (\phi_{\zeta_n-s}(-\theta) - 1) ds \\ &= \theta (1 - \epsilon) n + \frac{\mu}{n^\alpha} \int_0^{\zeta_n} n e^{\lambda_0 s} \left(\frac{e^{-\theta} e^{-\lambda_1 (\zeta_n-s)}}{1 - (1 - e^{-\lambda_1 (\zeta_n-s)}) e^{-\theta}} - 1 \right) ds. \end{aligned}$$

Let $\theta = \delta e^{-\lambda_1 \zeta_n}$, we can obtain

$$\begin{aligned} & n^{\alpha-1} \left(\theta (1 - \epsilon) n + \frac{\mu}{n^\alpha} \int_0^{\zeta_n} n e^{\lambda_0 s} \left(\frac{e^{-\theta} e^{-\lambda_1 (\zeta_n-s)}}{1 - (1 - e^{-\lambda_1 (\zeta_n-s)}) e^{-\theta}} - 1 \right) ds \right) \\ &= n^{\alpha-1} \delta e^{-\lambda_1 \zeta_n} (1 - \epsilon) n + n^{\alpha-1} \frac{\mu}{n^\alpha} \int_0^{\zeta_n} n e^{\lambda_0 s} \left(\frac{e^{-\delta e^{-\lambda_1 \zeta_n}} e^{-\lambda_1 (\zeta_n-s)}}{1 - (1 - e^{-\lambda_1 (\zeta_n-s)}) e^{-\delta e^{-\lambda_1 \zeta_n}}} - 1 \right) ds \\ &\rightarrow \delta \frac{\mu}{\lambda_1 - \lambda_0} (1 - \epsilon) - \mu \int_0^\infty e^{\lambda_0 s} \frac{\delta}{\delta + e^{\lambda_1 s}} ds \\ &= \delta \mu \left(\frac{1 - \epsilon}{\lambda_1 - \lambda_0} - \int_0^\infty \frac{e^{\lambda_0 s}}{\delta + e^{\lambda_1 s}} ds \right) \\ &= \delta \mu \left(\int_0^\infty \frac{(1 - \epsilon) e^{\lambda_0 s}}{e^{\lambda_1 s}} ds - \int_0^\infty \frac{e^{\lambda_0 s}}{\delta + e^{\lambda_1 s}} ds \right). \end{aligned}$$

By comparing the integrand, we have that

$$\begin{aligned} \frac{1 - \epsilon}{e^{\lambda_1 s}} - \frac{1}{\delta + e^{\lambda_1 s}} &< 0 \Leftrightarrow (1 - \epsilon) \delta - \epsilon e^{\lambda_1 s} < 0 \\ &\Leftrightarrow \delta < \frac{\epsilon e^{\lambda_1 s}}{1 - \epsilon} \\ &\Leftarrow \delta < \frac{\epsilon}{1 - \epsilon}. \end{aligned}$$

Hence, we conclude that for some $c > 0$

$$\limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P} \left(Z_1^n(\zeta_n) < (1-\epsilon)n \right) \leq -c.$$

Since $\mathbb{P}(\rho_n) \rightarrow 1$, the above result also holds when conditioned on the event ρ_n . Hence, we can obtain that for any $\epsilon \in (0, 1)$,

$$\begin{aligned} \mathbb{E}_{\rho_n} \left[\frac{n^4}{Z_1^n(\zeta_n)^4} \right] &= \mathbb{E}_{\rho_n} \left[\frac{n^4}{Z_1^n(\zeta_n)^4} \mid Z_1^n(\zeta_n) < (1-\epsilon)n \right] \mathbb{P}_{\rho_n} \left(Z_1^n(\zeta_n) < (1-\epsilon)n \right) \\ &\quad + \mathbb{E}_{\rho_n} \left[\frac{n^4}{Z_1^n(\zeta_n)^4} \mid Z_1^n(\zeta_n) \geq (1-\epsilon)n \right] \mathbb{P}_{\rho_n} \left(Z_1^n(\zeta_n) \geq (1-\epsilon)n \right) \\ &\leq n^4 \mathbb{P}_{\rho_n} \left(Z_1^n(\zeta_n) < (1-\epsilon)n \right) + \frac{1}{(1-\epsilon)^4} \\ &\rightarrow \frac{1}{(1-\epsilon)^4}. \end{aligned}$$

Since ϵ can be arbitrarily small, we have

$$\limsup_{n \rightarrow \infty} \mathbb{E}_{\rho_n} \left[\frac{n^4}{Z_1^n(\zeta_n)^4} \right] \leq 1.$$

By Jensen's inequality

$$\liminf_{n \rightarrow \infty} \mathbb{E}_{\rho_n} \left[\frac{n^4}{Z_1^n(\zeta_n)^4} \right] \geq \liminf_{n \rightarrow \infty} \frac{n^4}{(\mathbb{E}_{\rho_n} [Z_1^n(\zeta_n)])^4} = 1.$$

Similar results can be obtained for $\mathbb{E}_{\rho_n} \left[\frac{2n^2}{Z_1^n(\zeta_n)^2} \right]$. We conclude that

$$\mathbb{E}_{\rho_n} \left[\frac{n^4}{Z_1^n(\zeta_n)^4} - \frac{2n^2}{Z_1^n(\zeta_n)^2} + 1 \right] \rightarrow 0,$$

and hence

$$n^{1-\alpha} \mathbb{E}_{\rho_n} \left[\left| \tilde{R}_n - R_n \right| \right] \rightarrow 0. \quad (13)$$

The desired result follows by combining (13) and (11). $\square$

7 Proofs of Results in Section 4.1

7.1 Proof of Proposition 2

Proof Note that Proposition 2 does not rely on the assumption that sensitive cells have deterministic decay. Hence we provide the proof for the more general setting where $Z_0^n(t)$ is stochastic. We first observe that

$$\begin{aligned} & \mathbb{P}(I_n(\gamma_n) = k) \\ &= \mathbb{P}(I_n(\gamma_n) = k, I_n(\zeta_n) = k) + \mathbb{P}(I_n(\gamma_n) = k, I_n(\zeta_n) \neq k) \\ &= \mathbb{P}(I_n(\zeta_n) = k) \\ &\quad - \mathbb{P}(I_n(\gamma_n) \neq k, I_n(\zeta_n) = k) \\ &\quad + \mathbb{P}(I_n(\gamma_n) = k, I_n(\zeta_n) \neq k). \end{aligned} \tag{14}$$

For term (14),

$$\begin{aligned} \mathbb{P}(I_n(\gamma_n) \neq k, I_n(\zeta_n) = k) &= \mathbb{P}(I_n(\gamma_n) \neq k, I_n(\zeta_n) = k, \gamma_n \notin (\zeta_n - \delta, \zeta_n + \delta)) \\ &\quad + \mathbb{P}(I_n(\gamma_n) \neq k, I_n(\zeta_n) = k, \gamma_n \in (\zeta_n - \delta, \zeta_n + \delta)), \end{aligned}$$

where $\delta > 0$. For the first probability, by Theorem 1 of Hanagal et al. (2022), there exists $N_1 > 0$ such that when $n \geq N_1$,

$$\begin{aligned} & \sum_{k=1}^{\infty} \mathbb{P}(I_n(\gamma_n) \neq k, I_n(\zeta_n) = k, \gamma_n \notin (\zeta_n - \delta, \zeta_n + \delta)) \\ & \leq \mathbb{P}(\gamma_n \notin (\zeta_n - \delta, \zeta_n + \delta)) \\ & \leq \frac{\epsilon}{4}. \end{aligned}$$

For the second probability,

$$\begin{aligned} & \sum_{k=1}^{\infty} \mathbb{P}(I_n(\gamma_n) \neq k, I_n(\zeta_n) = k, \gamma_n \in (\zeta_n - \delta, \zeta_n + \delta)) \\ & \leq \mathbb{P}(I_n(\gamma_n) \neq I_n(\zeta_n), \gamma_n \in (\zeta_n - \delta, \zeta_n + \delta)) \\ & \leq \mathbb{P}(\exists t \in (\zeta_n - \delta, \zeta_n + \delta) \text{ such that } I_n(t) \neq I_n(\zeta_n), \gamma_n \in (\zeta_n - \delta, \zeta_n + \delta)) \\ & \leq \mathbb{P}(\text{mutation occurs in } (\zeta_n - \delta, \zeta_n + \delta)). \end{aligned}$$

The expected number of mutations between times $\zeta_n - \delta$ and $\zeta_n + \delta$ can be written as

$$\begin{aligned} \int_{\zeta_n - \delta}^{\zeta_n + \delta} \mu n^{-\alpha} \mathbb{E}[Z_0^n(t)] dt & \leq 2\delta \cdot \mu n^{-\alpha} \cdot n e^{\lambda_0(\zeta_n - \delta)} \\ & \leq C n^{1-\alpha + \frac{\lambda_0}{\lambda_1} \alpha}, \end{aligned}$$

where C is a constant. Then by Markov's Inequality and the fact that $\alpha > \frac{\lambda_1}{\lambda_1 - \lambda_0}$, there exists $N_2 > 0$ such that when $n \geq N_2$,

$$\mathbb{P}(\text{mutation occurs in } (\zeta_n - \delta, \zeta_n + \delta)) \leq \frac{\epsilon}{4}$$

Therefore, when $n \geq \max(N_1, N_2)$, $\sum_{k=1}^{\infty} \mathbb{P}(I_n(\gamma_n) \neq k, I_n(\zeta_n) = k) \leq \frac{\epsilon}{2}$. We can show a similar result for $\sum_{k=1}^{\infty} \mathbb{P}(I_n(\gamma_n) = k, I_n(\zeta_n) \neq k)$ with the same reasoning which completes the proof. $\square$

Prior to presenting the proof of Proposition 3, it is necessary to establish several preliminary results. We first consider the distribution of $I_n(\gamma_n)$ conditioned on the event of early recurrence. We show under assumption on α that $I_n(\gamma_n)$ is close to $I_n(\zeta_n - y)$ in total variation distance conditioned on the event of early recurrence.

Proposition 5 *If $\alpha \in \left(\frac{\lambda_1}{\lambda_1 - \lambda_0} \vee \frac{1}{2}, 1\right)$, then for all $y > 0$,*

$$\lim_{n \rightarrow \infty} TV(I_n(\gamma_n) | \gamma_n < \zeta_n - y, I_n(\zeta_n - y) | \gamma_n < \zeta_n - y) = 0.$$

Note that via Bayes rule we can write

$$\mathbb{P}(I_n(\zeta_n - y) = k | \gamma_n < \zeta_n - y) = Q_{k,n} \mathbb{P}(I_n(\zeta_n - y) = k),$$

where

$$Q_{k,n} = \frac{\mathbb{P}(\gamma_n < \zeta_n - y | I_n(\zeta_n - y) = k)}{\mathbb{P}(\gamma_n < \zeta_n - y)}.$$

By stochastic dominance, we can show that $Q_{k,n}$ increases in k . We then obtain the following result.

Lemma 5 *For all $k \in \mathbb{N}$,*

$$\mathbb{P}(I_n(\zeta_n - y) \geq k | \gamma_n < \zeta_n - y) \geq \mathbb{P}(I_n(\zeta_n - y) \geq k).$$

This result tells us that the number of clones at $\zeta_n - y$ conditioned on early recurrence stochastically dominates that without conditioning.

7.2 Proof of Proposition 5

Note that Proposition 5 does not rely on the assumption that sensitive cells have deterministic decay. Hence we provide the proof for the more general setting where $Z_0^n(t)$ is stochastic. We first show the following lemma.

Lemma 6 *Assume $\alpha \in \left(\frac{\lambda_1}{\lambda_1 - \lambda_0}, 1\right)$. For any $M > 0$, $y > 0$, and $\delta > 0$,*

$$\limsup_{n \rightarrow \infty} \frac{1}{n^\alpha} \log \mathbb{P} \left(\sup_{s \in (\zeta_n - y - \delta, \zeta_n - y)} Z_0^n(s) > Mn^\alpha \right) < 0.$$

Proof First, we observe that Lemma 6 follows directly if $Z_0^n(s) = ne^{-\lambda_0 s}$. From (1) and (2), we know that for a fixed $0 < \theta < \log\left(\frac{d_0}{r_0}\right)$, $\mathbb{E}[e^{\theta Z_0^n(s)}]$ exists for $s \in (0, \zeta_n)$. We first show that for $M_1 > 0$, $\limsup_{n \rightarrow \infty} \frac{1}{n^\alpha} \log \mathbb{P}(Z_0^n(\zeta_n - y - \delta) \geq M_1 n^\alpha) < 0$. We observe that

$$\begin{aligned} & \frac{1}{n^\alpha} \log \mathbb{P}(Z_0^n(\zeta_n - y - \delta) \geq M_1 n^\alpha) \\ &= \frac{1}{n^\alpha} \log \mathbb{P}\left(e^{\theta Z_0^n(\zeta_n - y - \delta)} \geq e^{M_1 \theta n^\alpha}\right) \\ &\leq \frac{1}{n^\alpha} \log \frac{\mathbb{E}[e^{\theta Z_0^n(\zeta_n - y - \delta)}]}{e^{M_1 \theta n^\alpha}} \\ &= n^{1-\alpha} \log \left(\frac{d_0(e^\theta - 1) - e^{-\lambda_0(\zeta_n - y - \delta)}(r_0 e^\theta - d_0)}{r_0(e^\theta - 1) - e^{-\lambda_0(\zeta_n - y - \delta)}(r_0 e^\theta - d_0)} \right) - M_1 \theta. \end{aligned}$$

By 3, we can show that

$$n^{1-\alpha} \log \left(\frac{d_0(e^\theta - 1) - e^{-\lambda_0(\zeta_n - y - \delta)}(r_0 e^\theta - d_0)}{r_0(e^\theta - 1) - e^{-\lambda_0(\zeta_n - y - \delta)}(r_0 e^\theta - d_0)} \right) = O\left(n^{1+\frac{\lambda_0 \alpha}{\lambda_1} - \alpha}\right).$$

Since $\alpha > \frac{\lambda_1}{\lambda_1 - \lambda_0}$, we have $\limsup_{n \rightarrow \infty} \frac{1}{n^\alpha} \log \mathbb{P}(Z_0^n(\zeta_n - y - \delta) \geq M_1 n^\alpha) < 0$. Next, we observe that for $M_2 > M_1$, we have

$$\begin{aligned} & \frac{1}{n^\alpha} \log \mathbb{P}\left(\sup_{s \in (\zeta_n - y - \delta, \zeta_n - y)} Z_0^n(s) > M_2 n^\alpha \mid Z_0^n(\zeta_n - y - \delta) < M_1 n^\alpha\right) \\ &\leq \frac{1}{n^\alpha} \log \frac{1 - (d_0/r_0)^{M_1 n^\alpha}}{1 - (d_0/r_0)^{M_2 n^\alpha}} \\ &\rightarrow (M_1 - M_2) \log\left(\frac{d_0}{r_0}\right) \\ &< 0, \end{aligned}$$

where we apply a gambler's ruin argument to obtain the first inequality. We then observe that

$$\begin{aligned} & \mathbb{P}\left(\sup_{s \in (\zeta_n - y - \delta, \zeta_n - y)} Z_0^n(s) > M_2 n^\alpha\right) \\ &= \mathbb{P}(Z_0^n(\zeta_n - y - \delta) \geq M_1 n^\alpha) \mathbb{P} \\ &\quad \left(\sup_{s \in (\zeta_n - y - \delta, \zeta_n - y)} Z_0^n(s) > M_2 n^\alpha \mid Z_0^n(\zeta_n - y - \delta) \geq M_1 n^\alpha\right) \\ &\quad + \mathbb{P}(Z_0^n(\zeta_n - y - \delta) < M_1 n^\alpha) \mathbb{P} \end{aligned}$$

$$\left(\sup_{s \in (\zeta_n - y - \delta, \zeta_n - y)} Z_0^n(s) > M_2 n^\alpha \mid Z_0^n(\zeta_n - y - \delta) < M_1 n^\alpha \right).$$

The desired result follows by applying Lemma 1.2.15 in Dembo and Zeitouni (2009) which is restated in the following lemma.

Lemma 7 *Let M be a fixed integer. Then, for every $a_\epsilon^i \geq 0$,*

$$\limsup_{\epsilon \rightarrow 0} \epsilon \log \left(\sum_{i=1}^M a_\epsilon^i \right) = \max_{i \in \{1, \dots, M\}} \limsup_{\epsilon \rightarrow 0} \epsilon \log a_\epsilon^i.$$

□

Proof We know that one half the total variation distance stated in Proposition 5 can be bounded above by

$$\begin{aligned} & \mathbb{P}(I_n(\gamma_n) \neq I_n(\zeta_n - y) \mid \gamma_n < \zeta_n - y) \\ &= \frac{\mathbb{P}(I_n(\gamma_n) \neq I_n(\zeta_n - y), \gamma_n < \zeta_n - y)}{\mathbb{P}(\gamma_n < \zeta_n - y)} \\ &\leq \frac{\mathbb{P}(\gamma_n < \zeta_n - y - \delta)}{\mathbb{P}(\gamma_n < \zeta_n - y)} + \frac{\mathbb{P}(I_n(\gamma_n) \neq I_n(\zeta_n - y), \zeta_n - y - \delta < \gamma_n < \zeta_n - y)}{\mathbb{P}(\gamma_n < \zeta_n - y - \delta)}. \end{aligned}$$

The first term goes to zero by Theorem 1. The second term is bounded above by the conditional probability $\mathbb{P}(I_n(\gamma_n) \neq I_n(\zeta_n - y) \mid \gamma_n \in (\zeta_n - y - \delta, \zeta_n - y))$. By Lemma 6, Theorem 1, and the assumption that $\alpha > \frac{1}{2}$, we have

$$\mathbb{P}\left(I_n(\gamma_n) \neq I_n(\zeta_n - y), \sup_{s \in (\zeta_n - y - \delta, \zeta_n - y)} Z_0^n(s) > Mn^\alpha \mid \gamma_n \in (\zeta_n - y - \delta, \zeta_n - y)\right) \rightarrow 0.$$

We also have

$$\begin{aligned} & \mathbb{P}\left(I_n(\gamma_n) \neq I_n(\zeta_n - y), \sup_{s \in (\zeta_n - y - \delta, \zeta_n - y)} Z_0^n(s) \leq Mn^\alpha \mid \gamma_n \in (\zeta_n - y - \delta, \zeta_n - y)\right) \\ &\leq \mathbb{P}\left(I_n(\gamma_n) \neq I_n(\zeta_n - y) \mid \sup_{s \in (\zeta_n - y - \delta, \zeta_n - y)} Z_0^n(s) \leq Mn^\alpha, \gamma_n \in (\zeta_n - y - \delta, \zeta_n - y)\right) \\ &\leq \mathbb{P}(\text{mutation occurs during } (\zeta_n - y - \delta, \zeta_n - y) \mid Z_0^n(s) = Mn^\alpha \forall s \in (\zeta_n - y - \delta, \zeta_n - y)) \\ &= 1 - e^{-\mu M \delta}. \end{aligned}$$

Since δ can be arbitrarily small, the desired result follows. □

7.3 Proof of Lemma 5

Proof We first show that $Q_{k,n}$ increases in k . If we define $Z_{1,k}^n(t)$ as the number of mutants at time t conditioned on the event of $\{I_n(\zeta_n - y) = k\}$, then it is easy to find a

coupling for $Z_{1,k}^n(t)$ and $Z_{1,k+1}^n(t)$ such that for $t \in (0, \zeta_n - y)$, $Z_{1,k+1}^n(t) \geq Z_{1,k}^n(t)$, which indicates that

$$\mathbb{P}(\gamma_n < \zeta_n - y | I_n(\zeta_n - y) = k + 1) \geq \mathbb{P}(\gamma_n < \zeta_n - y | I_n(\zeta_n - y) = k).$$

It then follows that $Q_{k,n}$ increases in k . Since

$$\mathbb{P}(I_n(\zeta_n - y) = k | \gamma_n < \zeta_n - y) = Q_{k,n} \mathbb{P}(I_n(\zeta_n - y) = k), \quad (16)$$

we now claim that $\sum_{k=0}^K \mathbb{P}(I_n(\zeta_n - y) = k | \gamma_n < \zeta_n - y) \leq \sum_{k=0}^K \mathbb{P}(I_n(\zeta_n - y) = k)$ for any non-negative integer K (which is equivalent to Lemma 5). Suppose (for a contradiction) that there exists $K > 0$ such that

$$\sum_{k=0}^K \mathbb{P}(I_n(\zeta_n - y) = k | \gamma_n < \zeta_n - y) > \sum_{k=0}^K \mathbb{P}(I_n(\zeta_n - y) = k). \quad (17)$$

Then we must have $Q_{K,n} > 1$ due to (16) and the fact that $Q_{k,n}$ increases in k . Moreover, for any $k \geq K$, $Q_{k,n} > 1$, which indicates that $\mathbb{P}(I_n(\zeta_n - y) = k | \gamma_n < \zeta_n - y) > \mathbb{P}(I_n(\zeta_n - y) = k)$ for any $k \geq K$ by (16). Combining this result with (17), we can obtain that $\sum_{k=0}^{\infty} \mathbb{P}(I_n(\zeta_n - y) = k | \gamma_n < \zeta_n - y) > \sum_{k=0}^{\infty} \mathbb{P}(I_n(\zeta_n - y) = k) = 1$ (a contradiction). The desired result then follows. $\square$

By utilizing Lemma 5, Proposition 2 and Proposition 5, we are able to prove Proposition 3.

7.4 Proof of Proposition 3

Proof Given $x > 0$, we have

$$\begin{aligned} & \mathbb{P}(I_n(\gamma_n) \geq x) - \mathbb{P}(I_n(\gamma_n) \geq x | \gamma_n < \zeta_n - y) \\ &= \mathbb{P}(I_n(\gamma_n) \geq x) - \mathbb{P}(I_n(\zeta_n - y) \geq x) \\ & \quad + \mathbb{P}(I_n(\zeta_n - y) \geq x | \gamma_n < \zeta_n - y) - \mathbb{P}(I_n(\gamma_n) \geq x | \gamma_n < \zeta_n - y) \\ & \quad + \mathbb{P}(I_n(\zeta_n - y) \geq x) - \mathbb{P}(I_n(\zeta_n - y) \geq x | \gamma_n < \zeta_n - y) \\ & \leq TV(I_n(\gamma_n), I_n(\zeta_n)) + TV(I_n(\zeta_n - y), I_n(\zeta_n)) + TV(I_n(\gamma_n) | \gamma_n \\ & \quad < \zeta_n - y, I_n(\zeta_n - y) | \gamma_n < \zeta_n - y) \\ & \quad + \mathbb{P}(I_n(\zeta_n - y) \geq x) - \mathbb{P}(I_n(\zeta_n - y) \geq x | \gamma_n < \zeta_n - y). \end{aligned}$$

We know from Lemma 5 that for all x ,

$$\mathbb{P}(I_n(\zeta_n - y) \geq x | \gamma_n < \zeta_n - y) \geq \mathbb{P}(I_n(\zeta_n - y) \geq x).$$

By Proposition 2 and Proposition 5, we have

$$\lim_{n \rightarrow \infty} TV(I_n(\gamma_n), I_n(\zeta_n)) = 0, \text{ and}$$

$$\lim_{n \rightarrow \infty} TV(I_n(\gamma_n) | \gamma_n < \zeta_n - y, I_n(\zeta_n - y) | \gamma_n < \zeta_n - y) = 0.$$

Hence, it remains to show that

$$\lim_{n \rightarrow \infty} TV(I_n(\zeta_n - y), I_n(\zeta_n)) = 0. \quad (18)$$

We know that

$$TV(I_n(\zeta_n - y), I_n(\zeta_n)) \leq \mathbb{P}(\text{mutation occurs in } (\zeta_n - y, \zeta_n)).$$

By a similar argument to that in the proof of Lemma 6, we can show that

$$\mathbb{P}\left(\sup_{s \in (\zeta_n - y, \zeta_n)} Z_0^n(s) > Mn^{\alpha^*}\right)$$

decays exponentially fast for $\frac{\lambda_1}{\lambda_1 - \lambda_0} < \alpha^* < \alpha$, and the desired result follows. $\square$

7.5 Proof of Lemma 2

We omit the detailed proof here as Lemma 9 (see Section 8) implies Lemma 2 with the additional argument that $\{\frac{I^n(\zeta_n - y)}{n^{1-\alpha}}\}$, conditioned on the event $A_{n,y}$, is uniformly integrable due to the exponential decay of its tail probability.

7.6 Proof of Lemma 3

Proof For $0 < \delta_1 < \delta_2$,

$$\mathbb{P}_{A_{n,y}}\left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta_1, 1 + \delta_2)\right) = \frac{\mathbb{P}\left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta_1, 1 + \delta_2), A_{n,y}\right)}{\mathbb{P}(A_{n,y})}. \quad (19)$$

We have the following upper and lower bound for the numerator in (19):

$$\mathbb{P}\left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta_1, 1 + \delta_2), A_{n,y}\right)$$

$$\leq \mathbb{P}\left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta_1, 1 + \delta_2), B_{n,y,(0,\zeta_n - y)} - B_{n,y,(t_1,t_2)} > n - (1 + \delta_2) \bar{B}_{n,y,(t_1,t_2)}\right), \quad (20)$$

and

$$\begin{aligned} & \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta_1, 1 + \delta_2), A_{n,y} \right) \\ & \geq \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta_1, 1 + \delta_2), B_{n,y,(0,\zeta_n-y)} - B_{n,y,(t_1,t_2)} > n - (1 + \delta_1) \bar{B}_{n,y,(t_1,t_2)} \right). \quad (21) \end{aligned}$$

By the Gartner-Ellis Theorem (Hollander 2000) and a similar calculation to that in the proof of Theorem 2 in Hanagal et al. (2022) (calculation of the moment generating function of $B_{n,y,(t_1,t_2)}$), we can obtain the large deviations rate for the probability presented in (21) by considering the following large deviation rates for $\delta, \varepsilon > 0$:

$$\begin{aligned} L_1(\delta) & \triangleq - \lim_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta, 1 + \delta + \varepsilon) \right) \\ & = \sup_{\theta \in (0,1)} \left[\mu \theta (1 + \delta) \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds - \mu \int_{t_1}^{t_2} \frac{\theta}{e^{\lambda_1 s} - \theta} e^{\lambda_0 s} ds \right], \end{aligned}$$

and

$$L_2(\delta) \triangleq - \lim_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P} (B_{n,y,(0,\zeta_n-y)} - B_{n,y,(t_1,t_2)} > n - (1 + \delta) \bar{B}_{n,y,(t_1,t_2)}) \quad (22)$$

$$\begin{aligned} & = \sup_{\theta \in (0,1)} \left[\frac{\mu \theta e^{\lambda_1 y}}{\lambda_1 - \lambda_0} - \mu \int_0^{t_1} \frac{\theta}{e^{\lambda_1 s} - \theta} e^{\lambda_0 s} ds - \mu \int_{t_2}^{\infty} \frac{\theta}{e^{\lambda_1 s} - \theta} e^{\lambda_0 s} ds \right. \\ & \quad \left. - \mu \theta (1 + \delta) \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds \right]. \quad (23) \end{aligned}$$

Since the generation of clones in different non-overlapping time periods are independent (the evolution of each clone is also independent), the large deviations rate for the probability presented in (21) is given by

$$L_1(\delta_1) + L_2(\delta_1).$$

We then restate the Envelope Theorem (see page 158 of Border (2020)) which can be used to analyze $L_1(\delta_1) + L_2(\delta_1)$.

Theorem 3 (Envelope Theorem) *Let X be a metric space and P an open subset of $\mathbb{R}^n$. Let $w : X \times P \rightarrow \mathbb{R}$ and assume $\frac{\partial w}{\partial p}$ exists and is continuous in $X \times P$. For each $p \in P$, let $x^*(p)$ maximize $w(x, p)$ over X . Set*

$$V(p) = w(x^*(p), p).$$

Assume that $x^* : P \rightarrow X$ is a continuous function. Then V is continuously differentiable and

$$DV(p) = \frac{\partial w(x, p)}{\partial p},$$

where the derivative is evaluated at the point $(x^*(p), p)$.

By the Envelope Theorem, we can obtain that $L_1(\delta)$ is convex, increasing in δ , and $L_2(\delta)$ is convex, decreasing in δ . Hence, by standard convex analysis, we can obtain the optimal δ^* to the following optimization problem:

$$\min_{\delta > 0} (L_1(\delta) + L_2(\delta)). \quad (24)$$

In particular,

$$\begin{aligned} \frac{dL_1(\delta)}{d\delta} &= \mu\theta_{1,\delta} \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds, \text{ and} \\ \frac{dL_2(\delta)}{d\delta} &= -\mu\theta_{2,\delta} \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds, \end{aligned}$$

where $\theta_{1,\delta}$ ($\theta_{2,\delta}$) is the maximizer to the optimization problem within the expression of $L_1(\delta)$ ($L_2(\delta)$). Therefore, the minimum of $(L_1(\delta) + L_2(\delta))$ is achieved when $\theta_{1,\delta} = \theta_{2,\delta}$. By some calculation, we can obtain that $\theta_{1,\delta^*} = \theta_{2,\delta^*} = \theta_y^*$, where δ^* is given by

$$\delta^* = \frac{\int_{t_1}^{t_2} \frac{e^{\lambda_1 s}}{(e^{\lambda_1 s} - \theta_y^*)^2} e^{\lambda_0 s} ds}{\int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds} - 1,$$

and θ_y^* is defined in (4). From (21), we know that

$$\begin{aligned} & \liminf_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P} \left(\frac{B_{n,y}(t_1, t_2)}{\bar{B}_{n,y}(t_1, t_2)} \in (1 + \delta^* - \epsilon, 1 + \delta^* + \epsilon), A_{n,y} \right) \\ & \geq \liminf_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P} \left(\frac{B_{n,y}(t_1, t_2)}{\bar{B}_{n,y}(t_1, t_2)} \in (1 + \delta^*, 1 + \delta^* + \epsilon), A_{n,y} \right) \\ & \geq - (L_1(\delta^*) + L_2(\delta^*)). \end{aligned}$$

To obtain the desired result, it suffices to show that

$$\begin{aligned} & \limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P} \left(\frac{B_{n,y}(t_1, t_2)}{\bar{B}_{n,y}(t_1, t_2)} \notin (1 + \delta^* - \epsilon, 1 + \delta^* + \epsilon), A_{n,y} \right) \\ & < - (L_1(\delta^*) + L_2(\delta^*)). \end{aligned}$$

We first investigate the event

$$\left\{ \frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta^* + \epsilon, \infty), A_{n,y} \right\}.$$

We note that this event can be decomposed into two parts:

$$\left\{ \frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in \left(1 + \delta^* + \epsilon, \frac{e^{\lambda_1 y}}{(\lambda_1 - \lambda_0) \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds} \right), A_{n,y} \right\},$$

where

$$\frac{e^{\lambda_1 y}}{(\lambda_1 - \lambda_0) \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds} = \lim_{n \rightarrow \infty} \frac{n}{\bar{B}_{n,y,(t_1,t_2)}},$$

and

$$\left\{ \frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in \left(\frac{e^{\lambda_1 y}}{(\lambda_1 - \lambda_0) \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds}, \infty \right), A_{n,y} \right\}.$$

We note that

$$\begin{aligned} & \limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in \left(\frac{e^{\lambda_1 y}}{(\lambda_1 - \lambda_0) \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds}, \infty \right), A_{n,y} \right) \\ & \leq \limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in \left(\frac{e^{\lambda_1 y}}{(\lambda_1 - \lambda_0) \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds}, \infty \right) \right) \\ & \leq -L_1 \left(\frac{e^{\lambda_1 y}}{(\lambda_1 - \lambda_0) \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds} - 1 \right) - L_2 \left(\frac{e^{\lambda_1 y}}{(\lambda_1 - \lambda_0) \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds} - 1 \right) \\ & < - (L_1(\delta^*) + L_2(\delta^*)), \end{aligned}$$

where the second inequality is due to the fact that

$$L_2 \left(\frac{e^{\lambda_1 y}}{(\lambda_1 - \lambda_0) \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds} - 1 \right) = 0,$$

and the last inequality is because δ^* is the optimal solution to (24). Therefore, we could safely omit the event

$$\left\{ \frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in \left(\frac{e^{\lambda_1 y}}{(\lambda_1 - \lambda_0) \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds}, \infty \right), A_{n,y} \right\},$$

and focus on the interval

$$\left(1 + \delta^* + \epsilon, \frac{e^{\lambda_1 y}}{(\lambda_1 - \lambda_0) \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds}\right).$$

To obtain the desired result, we need to utilize the upper bound introduced in 20. However, we cannot use it immediately as it does not follow the large deviation rate of 21. Hence, for an arbitrarily large but fixed integer M , we divide the interval

$$\left(1 + \delta^* + \epsilon, \frac{e^{\lambda_1 y}}{(\lambda_1 - \lambda_0) \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds}\right)$$

into M sub-intervals with equal length δ_M . Since M is a fixed number, we could analyze the event that $\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}}$ falls in each sub-interval, and then apply Lemma 7. By Lemma 7, to obtain the desired result, it suffices to show that for all $i \in \{0, \dots, M-1\}$,

$$\limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta^* + \epsilon + i\delta_M, 1 + \delta^* + \epsilon + (i+1)\delta_M), A_{n,y} \right) < -(L_1(\delta^*) + L_2(\delta^*)). \quad (25)$$

From (20), it suffices to show that

$$L_1(\delta^* + \epsilon + i\delta_M) + L_2(\delta^* + \epsilon + (i+1)\delta_M) > L_1(\delta^*) + L_2(\delta^*), \quad (26)$$

Since $L_2(\delta)$ is continuous for

$$\delta \in \left[\delta^* + \epsilon, \frac{e^{\lambda_1 y}}{(\lambda_1 - \lambda_0) \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds} - 1 \right]$$

and thus also uniform continuous, and δ^* is the optimal solution to (24), we conclude that for sufficiently large M , (26) holds for all $i \in \{0, \dots, M-1\}$ which validates (25).

By a similar argument, we can deal with the event

$$\left\{ \frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (0, 1 + \delta^* - \epsilon), A_{n,y} \right\},$$

which completes the proof. $\square$

7.7 Proof of Lemma 4

Proof We first notice that for $0 < \kappa_1 < \kappa_2 < \delta_1 < \delta_2$,

$$\mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta_1, 1 + \delta_2), \frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1 + \kappa_1, 1 + \kappa_2) \right)$$

$$\begin{aligned}
&\leq \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta_1, \infty), \frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1 + \kappa_1, 1 + \kappa_2) \right) \\
&\leq \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta_1, \infty) \middle| I_{n,(t_1,t_2)} = \lfloor (1 + \kappa_2) \bar{I}_{n,(t_1,t_2)} \rfloor \right) \\
&\quad \times \mathbb{P} \left(\frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1 + \kappa_1, 1 + \kappa_2) \right), \tag{27}
\end{aligned}$$

where the last inequality is due to stochastic dominance (an increased number of clones leads to a larger population of mutants). With a similar reasoning, we also have

$$\begin{aligned}
&\mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta_1, 1 + \delta_2), \frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1 + \kappa_1, 1 + \kappa_2) \right) \\
&= \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta_1, \infty), \frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1 + \kappa_1, 1 + \kappa_2) \right) \\
&\quad - \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta_2, \infty), \frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1 + \kappa_1, 1 + \kappa_2) \right) \\
&\geq \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta_1, \infty) \middle| I_{n,(t_1,t_2)} = \lfloor (1 + \kappa_1) \bar{I}_{n,(t_1,t_2)} \rfloor \right) \\
&\quad \times \mathbb{P} \left(\frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1 + \kappa_1, 1 + \kappa_2) \right) \\
&\quad - \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta_2, \infty) \middle| I_{n,(t_1,t_2)} = \lfloor (1 + \kappa_2) \bar{I}_{n,(t_1,t_2)} \rfloor \right) \\
&\quad \times \mathbb{P} \left(\frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1 + \kappa_1, 1 + \kappa_2) \right). \tag{28}
\end{aligned}$$

By considering the sum of $\lfloor (1 + \kappa_1) \bar{I}_{n,(t_1,t_2)} \rfloor$ i.i.d. random variables, each of which is the number of descendants of one mutated cell, we can compute (using the Cramér's theorem) that

$$\begin{aligned}
&L_1(\delta_1, \kappa_1) \\
&= - \lim_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta_1, 1 + \delta_2) \middle| I_{n,(t_1,t_2)} = \lfloor (1 + \kappa_1) \bar{I}_{n,(t_1,t_2)} \rfloor \right) \\
&= \sup_{\theta \in (0,1)} \left[\mu \theta (1 + \delta_1) \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds - \mu (1 + \kappa_1) \int_{t_1}^{t_2} e^{\lambda_0 s} ds \log \left(\frac{\int_{t_1}^{t_2} \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta} e^{\lambda_0 s} ds}{\int_{t_1}^{t_2} e^{\lambda_0 s} ds} \right) \right].
\end{aligned}$$

Because the distribution of $I_{n,(t_1,t_2)}$ is Poisson with a mean of $\bar{I}_{n,(t_1,t_2)}$, given in (6), we can compute that

$$\begin{aligned}
L_2(\kappa_1) &= - \lim_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P} \left(\frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1 + \kappa_1, 1 + \kappa_2) \right) \\
&= \mu \int_{t_1}^{t_2} e^{\lambda_0 s} ds ((1 + \kappa_1) \log(1 + \kappa_1) - (1 + \kappa_1) + 1).
\end{aligned}$$

Consider the optimization problem:

$$\min_{\kappa > 0} (L_1(\delta^*, \kappa) + L_2(\kappa)). \quad (29)$$

By the Envelope Theorem, we can show that $L_1(\delta^*, \kappa)$ is convex, decreasing in κ and $L_2(\kappa)$ is convex, increasing in κ . Hence, by standard convex analysis, if $t_2 - t_1 < -\frac{1}{\lambda_1} \log\left(\frac{1}{2 - \theta_y^*}\right)$ (this condition guarantees that the optimal solution lies in $(0, \delta^*)$), the optimal value κ^* is given by

$$\kappa^* = \frac{\int_{t_1}^{t_2} \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta_y^*} e^{\lambda_0 s} ds}{\int_{t_1}^{t_2} e^{\lambda_0 s} ds} - 1,$$

where θ_y^* is defined in (4).

By a similar argument to that in the proof of Lemma 3, we can focus on the event

$$\left\{ \frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta^* - \epsilon, 1 + \delta^* + \epsilon), \frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1, 1 + \delta^* - \epsilon) \right\}.$$

From (28), we have

$$\begin{aligned} & \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta^* - \epsilon, 1 + \delta^* + \epsilon), \frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma) \right) \quad (30) \\ & \geq \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta^* - \epsilon, \infty) \middle| I_{n,(t_1,t_2)} = \lfloor (1 + \kappa^* - \sigma) \bar{I}_{n,(t_1,t_2)} \rfloor \right) \\ & \quad \times \mathbb{P} \left(\frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma) \right) \\ & - \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta^* + \epsilon, \infty) \middle| I_{n,(t_1,t_2)} = \lfloor (1 + \kappa^* + \sigma) \bar{I}_{n,(t_1,t_2)} \rfloor \right) \\ & \quad \times \mathbb{P} \left(\frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma) \right). \end{aligned}$$

To use this lower bound, we need to make sure that the negative part decays exponentially faster. By the calculation of L_1 , we can obtain that for a positive number $N > \frac{\sigma(1+\delta^*)}{\epsilon(1+\kappa^*)}$, we have $\frac{1+\delta^*+\epsilon}{1+\delta^*-\epsilon} > \frac{1+\kappa^*+\frac{\sigma}{N}}{1+\kappa^*-\frac{\sigma}{N}}$ which indicates that $L_1(\delta^* - \epsilon, \kappa^* - \frac{\sigma}{N}) < L_1(\delta^* + \epsilon, \kappa^* + \frac{\sigma}{N})$. Therefore,

$$\begin{aligned} & \liminf_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta^* - \epsilon, 1 + \delta^* + \epsilon), \frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma) \right) \\ & \geq \liminf_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P} \left(\frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta^* - \epsilon, 1 + \delta^* + \epsilon), \frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in \left(1 + \kappa^* - \frac{\sigma}{N}, 1 + \kappa^* + \frac{\sigma}{N} \right) \right) \\ & \geq - \left(L_1 \left(\delta^* - \epsilon, \kappa^* - \frac{\sigma}{N} \right) + L_2 \left(\kappa^* - \frac{\sigma}{N} \right) \right). \end{aligned}$$

We first divide the interval $(0, \kappa^* - \sigma)$ into M sub-intervals with equal length δ_M for an arbitrarily large but fixed integer $M > 0$. By Lemma 7, to obtain the desired result, the first step is to show that for all $i \in \{0, \dots, M-1\}$,

$$\begin{aligned} & \limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P} \left(\frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1 + i\delta_M, 1 + (i+1)\delta_M), A_{n,y,\epsilon}^{t_1,t_2} \right) \\ & < - \left(L_1 \left(\delta^* - \epsilon, \kappa^* - \frac{\sigma}{N} \right) + L_2 \left(\kappa^* - \frac{\sigma}{N} \right) \right). \end{aligned}$$

From (27), it suffices to show that

$$L_1 \left(\delta^* - \epsilon, (i+1)\delta_M \right) + L_2(i\delta_M) > L_1 \left(\delta^* - \epsilon, \kappa^* - \frac{\sigma}{N} \right) + L_2 \left(\kappa^* - \frac{\sigma}{N} \right). \quad (31)$$

We first analyze the left hand side of (31). Since $L_1(\delta^*, \kappa)$ is continuous for $\kappa \in [0, \delta^* - \epsilon]$, and thus also uniformly continuous, we have

$$\lim_{M \rightarrow \infty} \sup_{i \in \{0, \dots, M-1\}} |L_1(\delta^*, i\delta_M) - L_1(\delta^*, (i+1)\delta_M)| = 0. \quad (32)$$

Let $\theta_y^*(\delta_1, \kappa_1)$ be the optimal value for the inner optimization problem appearing in the calculation of $L_1(\delta_1, \kappa_1)$. We can see that $\theta_y^*(\delta_1, \kappa_1)$ decreases in κ_1 . Hence, by the Envelope Theorem,

$$\frac{\partial L_1(\delta_1, \kappa_1)}{\partial \delta_1} = \mu \theta_y^*(\delta_1, \kappa_1) \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds \leq \mu \theta_y^*(\delta_1, 0) \int_{t_1}^{t_2} e^{-(\lambda_1 - \lambda_0)s} ds,$$

where the upper bound does not depend on κ_1 . Therefore,

$$\lim_{\epsilon \rightarrow 0} \sup_{\kappa \in (0, \delta^* - \epsilon)} |L_1(\delta^* - \epsilon, \kappa) - L_1(\delta^*, \kappa)| = 0. \quad (33)$$

By (32) and (33), we know that

$$\sup_{i \in \{0, \dots, M-1\}} |L_1(\delta^* - \epsilon, (i+1)\delta_M) + L_2(i\delta_M) - L_1(\delta^*, i\delta_M) - L_2(i\delta_M)|$$

can be arbitrarily small for sufficiently small ϵ and sufficiently large M . We then analyze the right hand side of (31). It is easy to see that

$$\left| L_1 \left(\delta^* - \epsilon, \kappa^* - \frac{\sigma}{N} \right) + L_2 \left(\kappa^* - \frac{\sigma}{N} \right) - L_1(\delta^*, \kappa^*) - L_2(\kappa^*) \right|$$

can be arbitrarily small for sufficiently small ϵ and sufficiently large N , which completes the first step which shows that the probability of the following event,

$$\left\{ \frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta^* - \epsilon, 1 + \delta^* + \epsilon), \frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1, 1 + \kappa^* - \sigma) \right\},$$

decays faster than that of the event of interest (the probability of which is evaluated in (30)).

By a similar argument, we can deal with the following event (the second step)

$$\left\{ \frac{B_{n,y,(t_1,t_2)}}{\bar{B}_{n,y,(t_1,t_2)}} \in (1 + \delta^* - \epsilon, 1 + \delta^* + \epsilon), \frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1 + \kappa^* + \sigma, 1 + \delta^* - \epsilon) \right\},$$

which completes the proof. $\square$

7.8 Proof of Proposition 4

Proof We first observe that

$$\begin{aligned} & \mathbb{P}_{A_{n,y}} \left(\frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \notin (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma) \right) \\ &= \mathbb{P}_{A_{n,y}} \left(\frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \notin (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma), A_{n,y,\epsilon}^{t_1,t_2,c} \right) \\ & \quad + \mathbb{P}_{A_{n,y}} \left(\frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \notin (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma), A_{n,y,\epsilon}^{t_1,t_2} \right). \end{aligned}$$

By Lemma 3, we know that the first probability decays exponentially fast, and thus we focus on the second probability. We have

$$\begin{aligned} & \mathbb{P}_{A_{n,y}} \left(\frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \notin (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma), A_{n,y,\epsilon}^{t_1,t_2} \right) \\ &= \frac{\mathbb{P} \left(\frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \notin (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma), A_{n,y,\epsilon}^{t_1,t_2} \right)}{\mathbb{P}(A_{n,y})} \\ & \quad \times \mathbb{P} \left(A_{n,y} \left| \frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \notin (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma), A_{n,y,\epsilon}^{t_1,t_2} \right. \right). \end{aligned}$$

Because

$$\begin{aligned} & \mathbb{P} \left(A_{n,y} \left| \frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \notin (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma), A_{n,y,\epsilon}^{t_1,t_2} \right. \right) \leq \mathbb{P}(A_{n,y} | B_{n,y,(t_1,t_2)}) \\ &= \lceil (1 + \delta^* + \epsilon) \bar{B}_{n,y,(t_1,t_2)} \rceil, \end{aligned}$$

and

$$\begin{aligned} & \mathbb{P} \left(A_{n,y} \left| \frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \notin (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma), A_{n,y,\epsilon}^{t_1,t_2} \right. \right) \geq \mathbb{P}(A_{n,y} | B_{n,y,(t_1,t_2)}) \\ &= \lfloor (1 + \delta^* - \epsilon) \bar{B}_{n,y,(t_1,t_2)} \rfloor, \end{aligned}$$

by Theorem 1, we have

$$\left| \limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \mathbb{P} \left(A_{n,y} \left| \frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \notin (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma), A_{n,y,\epsilon}^{t_1,t_2} \right. \right) \right. \\ \left. - \limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \mathbb{P} \left(A_{n,y} \left| A_{n,y,\epsilon}^{t_1,t_2} \right. \right) \right| \rightarrow 0$$

as ϵ goes to zero. From the proof of Lemma 4, we know that for sufficiently small ϵ , the large deviations rate of

$$\mathbb{P} \left(\frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \notin (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma), A_{n,y,\epsilon}^{t_1,t_2} \right)$$

is larger than that of

$$\mathbb{P} \left(\frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \in (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma), A_{n,y,\epsilon}^{t_1,t_2} \right),$$

and their difference is bounded away from zero. The desired result then follows from the fact that

$$\frac{\mathbb{P} \left(\frac{I_{n,(t_1,t_2)}}{\bar{I}_{n,(t_1,t_2)}} \notin (1 + \kappa^* - \sigma, 1 + \kappa^* + \sigma), A_{n,y,\epsilon}^{t_1,t_2} \right)}{\mathbb{P} (A_{n,y,\epsilon}^{t_1,t_2})} \frac{\mathbb{P} (A_{n,y,\epsilon}^{t_1,t_2})}{\mathbb{P} (A_{n,y})} \mathbb{P} (A_{n,y} | A_{n,y,\epsilon}^{t_1,t_2})$$

decays exponentially fast as the first term decays exponentially fast by Lemma 4 and $\frac{\mathbb{P} (A_{n,y,\epsilon}^{t_1,t_2})}{\mathbb{P} (A_{n,y})} \mathbb{P} (A_{n,y} | A_{n,y,\epsilon}^{t_1,t_2}) \approx 1$ by Lemma 3. $\square$

8 Proof of Results in Section 4.2

8.1 Proof of Theorem 2

Proof The key idea of the proof is to carefully choose a sequence of the most likely events $E_{n,y,\epsilon_1} \subset A_{n,y}$, such that

$$\frac{\mathbb{P} (A_{n,y} \setminus E_{n,y,\epsilon_1})}{\mathbb{P} (A_{n,y})}$$

decays exponentially fast. Then it suffices to calculate the Simpson's Index conditioned on E_{n,y,ϵ_1} . Our choice of E_{n,y,ϵ_1} will make the analysis much easier.

Recall that we denote by $I_n(\zeta_n - y)$ the number of clones generated in the time period $(0, \zeta_n - y)$. In Hanagal et al. (2022) (see Section 3.2), we obtain the most likely number of clones given early recurrence has occurred. We restate the result in the following lemma.

Lemma 8

$$\begin{aligned} & \operatorname{argmax}_{a \in (0, e^{\lambda_1 y}]} \lim_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P} \left(\gamma_n \leq \zeta_n - y \mid I_n(\zeta_n - y) = \lfloor -a \frac{\mu}{\lambda_0} n^{1-\alpha} \rfloor \right) \\ & \mathbb{P} \left(I_n(\zeta_n - y) = \lfloor -a \frac{\mu}{\lambda_0} n^{1-\alpha} \rfloor \right) \\ & = -\lambda_0 \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta_y^*} e^{\lambda_0 s} ds, \end{aligned}$$

where θ_y^* is defined in (4).

From Lemma 8, we know that given early recurrence has happened, the number of clones is approximately $\mu n^{1-\alpha} \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta_y^*} e^{\lambda_0 s} ds$. Hence we consider the following set of events for $\epsilon_1 > 0$:

$$E_{n,y,\epsilon_1} = \{\gamma_n < \zeta_n - y, I_n(\zeta_n - y) \in O_{n,y,\epsilon_1}\},$$

where

$$O_{n,y,\epsilon_1} = \left((1 - \epsilon_1) \mu n^{1-\alpha} \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta_y^*} e^{\lambda_0 s} ds, (1 + \epsilon_1) \mu n^{1-\alpha} \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta_y^*} e^{\lambda_0 s} ds \right).$$

We choose E_{n,y,ϵ_1} in this way so that the number of clones is concentrated. For simplicity, we let $\mathbb{P}_{E_{n,y,\epsilon_1}}(\cdot) = \mathbb{P}(\cdot | E_{n,y,\epsilon_1})$. We then analyze $\mathbb{P}(A_{n,y} \setminus E_{n,y,\epsilon_1})$ in the next lemma.

Lemma 9

$$\limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P}(A_{n,y} \setminus E_{n,y,\epsilon_1}) < -L(y),$$

where

$$L(y) \triangleq -\lim_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P}(A_{n,y}) = \sup_{\theta \in (0,1)} \left[\frac{\theta \mu e^{\lambda_1 y}}{\lambda_1 - \lambda_0} - \mu \theta \int_0^\infty \frac{e^{\lambda_0 s}}{e^{\lambda_1 s} - \theta} ds \right].$$

Proof See Section 8.1.1. □

By Lemma 9, $\frac{\mathbb{P}(A_{n,y} \setminus E_{n,y,\epsilon_1})}{\mathbb{P}(A_{n,y})}$ decays exponentially fast as desired. Since Simpson's Index is at most 1, we have

$$\lim_{n \rightarrow \infty} \left| n^{1-\alpha} \mathbb{E}_{A_{n,y}} [R_{n,y}] - n^{1-\alpha} \mathbb{E}_{E_{n,y,\epsilon_1}} [R_{n,y}] \right| = 0.$$

Notice that in the time period $(0, \zeta_n - y)$, each clone is generated according to a Poisson process. We denote by $X_{n,y,i}$ the size of the i -th clone at $\zeta_n - y$. Note that the mutant clones are ordered at random, not in chronological order by when the mutation occurred. For ease of exposition, we drop the subscript n and y . Since the generation

times of clones as well as their evolution are independent conditioned on the total number of clones generated in the time period $(0, \zeta_n - y)$, we can obtain that

$$\mathbb{E}[X_i] = e^{\lambda_1(\zeta_n - y)} \frac{\int_0^{\zeta_n - y} e^{-(\lambda_1 - \lambda_0)t} dt}{\int_0^{\zeta_n - y} e^{\lambda_0 t} dt}. \quad (34)$$

Hence, it is convenient to condition on the number of clones and then apply the law of total expectation:

$$\begin{aligned} n^{1-\alpha} \mathbb{E}_{E_{n,y,\epsilon_1}}[R_{n,y}] &= n^{1-\alpha} \sum_{k \in O_{n,y,\epsilon_1}} \mathbb{E}[R_{n,y} | \gamma_n < \zeta_n - y, I_n(\zeta_n - y) = k] \\ &= \frac{\mathbb{P}(\gamma_n < \zeta_n - y, I_n(\zeta_n - y) = k)}{\mathbb{P}(E_{n,y,\epsilon_1})}. \end{aligned}$$

Define $\tilde{R}_{n,y} = \frac{\sum_{i=1}^{I_n(\zeta_n - y)} X_i^2}{n^2}$. We have

$$\begin{aligned} & n^{1-\alpha} \mathbb{E}_{A_{n,y}} \left[\left| \tilde{R}_{n,y} - R_{n,y} \right| \right] \\ &= n^{1-\alpha} \mathbb{E}_{A_{n,y}} \left[\left| \sum_{i=1}^{I_n(\zeta_n - y)} \left(\frac{X_i}{n} \right)^2 - \sum_{i=1}^{I_n(\zeta_n - y)} \left(\frac{X_i}{Z_1^n(\zeta_n - y)} \right)^2 \right| \right] \\ &= n^{1-\alpha} \mathbb{E}_{A_{n,y}} \left[\left| \frac{\sum_{i=1}^{I_n(\zeta_n - y)} (X_i)^2}{n^2} \left(\frac{n^2}{Z_1^n(\zeta_n - y)^2} - 1 \right) \right| \right] \\ &= n^{1-\alpha} \mathbb{E}_{A_{n,y}} \left[\left| \frac{\sum_{i=1}^{I_n(\zeta_n - y)} (X_i)^2}{n^2} \left(\frac{n^2}{Z_1^n(\zeta_n - y)^2} - 1 \right) \right| \mathbb{1}_{Z_1^n(\zeta_n - y) < (1+\epsilon)n} \right] \quad (35) \\ &+ n^{1-\alpha} \mathbb{E}_{A_{n,y}} \left[\left| \frac{\sum_{i=1}^{I_n(\zeta_n - y)} (X_i)^2}{n^2} \left(\frac{n^2}{Z_1^n(\zeta_n - y)^2} - 1 \right) \right| \mathbb{1}_{Z_1^n(\zeta_n - y) \geq (1+\epsilon)n} \right]. \quad (36) \end{aligned}$$

Because $\mathbb{E}[\tilde{R}_{n,y} | \gamma_n < \zeta_n - y] = O(n^{\alpha-1})$ (the main result of this Section), 35 is upper bounded by $C\epsilon$ as n goes to infinity, where C is some constant. Moreover,

$$\begin{aligned} 36 &\leq n^{1-\alpha} \mathbb{E}_{A_{n,y}} \left[\frac{\sum_{i=1}^{I_n(\zeta_n - y)} (X_i)^2}{n^2} \mathbb{1}_{Z_1^n(\zeta_n - y) \geq (1+\epsilon)n} \right] \\ &\leq n^{1-\alpha} \mathbb{E}_{A_{n,y}} \left[\frac{(Z_1^n(\zeta_n - y))^2}{n^2} \mathbb{1}_{Z_1^n(\zeta_n - y) \geq (1+\epsilon)n} \right] \\ &= n^{1-\alpha} \mathbb{E} \left[\frac{(Z_1^n(\zeta_n - y))^2}{n^2} \mathbb{1}_{Z_1^n(\zeta_n - y) \geq (1+\epsilon)n} \right] / \mathbb{P}(Z_1^n(\zeta_n - y) \geq n) \end{aligned}$$

$$\leq n^{1-\alpha} \mathbb{E} \left[\frac{(Z_1^n(\zeta_n - y))^4}{n^4} \right]^{\frac{1}{2}} \frac{\mathbb{P}(Z_1^n(\zeta_n - y) \geq (1 + \epsilon)n)}{\mathbb{P}(Z_1^n(\zeta_n - y) \geq n)} \quad (37)$$

$$\rightarrow 0 \text{ as } n \text{ goes to infinity,} \quad (38)$$

where we use Cauchy-Schwarz inequality to get (37), and (38) is due to the fact⁵ that

$$\begin{aligned} & \lim_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P}(Z_1^n(\zeta_n - y) > (1 + \epsilon)n) \\ &= - \sup_{\theta \in (0,1)} \left[(1 + \epsilon) \frac{\theta \lambda_1 \mu e^{\lambda_1 y}}{r_1(\lambda_1 - \lambda_0)} - \frac{\mu \lambda_1}{r_1} \int_0^\infty \frac{\theta e^{-rs}}{e^{\lambda_1 s} - \theta} ds \right], \end{aligned}$$

and $\mathbb{E}[Z_1^n(\zeta_n - y)^4] \leq Cn^4$ for some constant $C > 0$ (see the proof on Page 51 of Leder et al. (2024)). Therefore

$$\lim_{n \rightarrow \infty} \left| n^{1-\alpha} \mathbb{E}[R_{n,y} | \gamma_n < \zeta_n - y] - n^{1-\alpha} \mathbb{E}[\tilde{R}_{n,y} | \gamma_n < \zeta_n - y] \right| = 0.$$

Hence, it suffices to analyze

$$\begin{aligned} & n^{1-\alpha} \mathbb{E} \left[\tilde{R}_{n,y} | \gamma_n < \zeta_n - y, I_n(\zeta_n - y) = k \right] \\ &= \frac{k}{n^{1+\alpha}} \mathbb{E} \left[X_1^2 \left| \sum_{i=1}^k X_i \geq n \right. \right] \\ &= \frac{ke^{2\lambda_1(\zeta_n - y)}}{n^{1+\alpha}} \mathbb{E} \left[\left(e^{-\lambda_1(\zeta_n - y)} X_1 \right)^2 \left| \sum_{i=1}^k e^{-\lambda_1(\zeta_n - y)} X_i \geq e^{-\lambda_1(\zeta_n - y)} n \right. \right]. \quad (39) \end{aligned}$$

The next step is to find a bound on

$$\mathbb{E} \left[\left(e^{-\lambda_1(\zeta_n - y)} X_1 \right)^2 \left| \sum_{i=1}^k e^{-\lambda_1(\zeta_n - y)} X_i \geq e^{-\lambda_1(\zeta_n - y)} n \right. \right], \quad (40)$$

for $k \in O_{n,y,\epsilon_1}$. We note that $\sum_{i=1}^k e^{-\lambda_1(\zeta_n - y)} X_i \geq e^{-\lambda_1(\zeta_n - y)} n$ is equivalent to $\frac{\sum_{i=1}^k e^{-\lambda_1(\zeta_n - y)} X_i}{k} \geq q_{k,n} \mathbb{E}[e^{-\lambda_1(\zeta_n - y)} X_i]$, where

$$q_{k,n} = \frac{n}{k \mathbb{E}[X_1]}.$$

Since

$$\frac{\lambda_1 - \lambda_0}{\mu} n^\alpha \leq e^{\lambda_1 \zeta_n} \leq \frac{\lambda_1 - \lambda_0}{\mu} n^\alpha + 1,$$

⁵ This result can be shown using analogous arguments to those employed in the proof of Theorem 1.

we have

$$\lim_{n \rightarrow \infty} \frac{n / \left((1 - \epsilon_1) \mu n^{1-\alpha} \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta_y^*} e^{\lambda_0 s} ds \right)}{\mathbb{E}[X_1]} = - \frac{e^{\lambda_1 y}}{(1 - \epsilon_1) \lambda_0 \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta_y^*} e^{\lambda_0 s} ds},$$

and

$$\lim_{n \rightarrow \infty} \frac{n / \left((1 + \epsilon_1) \mu n^{1-\alpha} \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta_y^*} e^{\lambda_0 s} ds \right)}{\mathbb{E}[X_1]} = - \frac{e^{\lambda_1 y}}{(1 + \epsilon_1) \lambda_0 \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta_y^*} e^{\lambda_0 s} ds}.$$

Hence, for arbitrarily small but fixed $\epsilon_2 > 0$, there exists $N_{\epsilon_2} > 0$ such that when $n > N_{\epsilon_2}$, $k \in O_{n,y,\epsilon_1}$ implies that

$$q_{k,n} \in \tilde{O}_{y,\epsilon_1,\epsilon_2} = \left[\frac{(1 - \epsilon_2)}{(1 + \epsilon_1)} q^*, \frac{(1 + \epsilon_2)}{(1 - \epsilon_1)} q^* \right]. \quad (41)$$

where

$$q^* = - \frac{e^{\lambda_1 y}}{\lambda_0 \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta_y^*} e^{\lambda_0 s} ds}. \quad (42)$$

Since $q^* > 1$ by the definition of θ_y^* (see Lemma 1 in Hanagal et al. (2022)), we will make ϵ_1, ϵ_2 sufficiently small such that $\frac{(1 - \epsilon_2)}{(1 + \epsilon_1)} q^* > 1$ in the rest of the proof. Let

$$\begin{aligned} Q_{k,n} &\triangleq q_{k,n} \mathbb{E} \left[e^{-\lambda_1 (\zeta_n - y)} X_1 \right] \\ &= q_{k,n} \frac{\int_0^{\zeta_n - y} e^{-(\lambda_1 - \lambda_0)t} dt}{\int_0^{\zeta_n - y} e^{\lambda_0 t} dt}, \end{aligned} \quad (43)$$

where the equality follows from (34). Note that $Q_{k,n}$ depends on y , but we omit this dependence for simplicity. Define

$$Q^* \triangleq \lim_{n \rightarrow \infty} q^* \frac{\int_0^{\zeta_n - y} e^{-(\lambda_1 - \lambda_0)t} dt}{\int_0^{\zeta_n - y} e^{\lambda_0 t} dt} = - \frac{\lambda_0}{\lambda_1 - \lambda_0} q^*.$$

We then calculate the log moment-generating function for $e^{-\lambda_1 (\zeta_n - y)} X_1$ and its derivatives. For $\theta \in (0, 1)$, let

$$\begin{aligned}
\Lambda_n(\theta) &\triangleq \log \mathbb{E} \left[e^{\theta e^{-\lambda_1(\zeta_n - y)} X_1} \right] \\
&= \log \left(\frac{\int_0^{\zeta_n - y} e^{\lambda_0 t} \frac{e^{\lambda_1 t} e^{-\lambda_1(\zeta_n - y)} e^{\theta e^{-\lambda_1(\zeta_n - y)}}}{e^{\lambda_1 t} e^{-\lambda_1(\zeta_n - y)} e^{\theta e^{-\lambda_1(\zeta_n - y)}} - e^{\theta e^{-\lambda_1(\zeta_n - y)}} + 1} dt}{\int_0^{\zeta_n - y} e^{\lambda_0 t} dt} \right) \\
&\rightarrow \log \left(-\lambda_0 \int_0^\infty e^{\lambda_0 t} \frac{1}{1 - \theta e^{-\lambda_1 t}} dt \right) \text{ as } n \rightarrow \infty.
\end{aligned}$$

We denote the limit by $\Lambda(\theta)$. Taking the first derivative, we have

$$\begin{aligned}
\Lambda'_n(\theta) &= \frac{\mathbb{E} \left[e^{-\lambda_1(\zeta_n - y)} X_i e^{\theta e^{-\lambda_1(\zeta_n - y)} X_1} \right]}{\mathbb{E} \left[e^{\theta e^{-\lambda_1(\zeta_n - y)} X_1} \right]} \\
&= \frac{\int_0^{\zeta_n - y} e^{\lambda_0 t} \frac{e^{\lambda_1 t} e^{-2\lambda_1(\zeta_n - y)} e^{-\theta e^{-\lambda_1(\zeta_n - y)}}}{\left(e^{\lambda_1 t} e^{-\lambda_1(\zeta_n - y)} - 1 + e^{-\theta e^{-\lambda_1(\zeta_n - y)}} \right)^2} dt}{\int_0^{\zeta_n - y} e^{\lambda_0 t} \frac{e^{\lambda_1 t} e^{-\lambda_1(\zeta_n - y)}}{e^{\lambda_1 t} e^{-\lambda_1(\zeta_n - y)} - 1 + e^{-\theta e^{-\lambda_1(\zeta_n - y)}}} dt} \\
&\rightarrow \frac{\int_0^\infty e^{\lambda_0 t} \frac{e^{-\lambda_1 t}}{(1 - \theta e^{-\lambda_1 t})^2} dt}{\int_0^\infty e^{\lambda_0 t} \frac{1}{1 - \theta e^{-\lambda_1 t}} dt} \text{ as } n \rightarrow \infty.
\end{aligned}$$

We denote the limit by $\Lambda'(\theta)$. Take the second derivative, we have

$$\begin{aligned}
\Lambda''_n(\theta) &= \frac{\mathbb{E} \left[e^{-2\lambda_1(\zeta_n - y)} X_1^2 e^{\theta e^{-\lambda_1(\zeta_n - y)} X_1} \right] \mathbb{E} \left[e^{\theta e^{-\lambda_1(\zeta_n - y)} X_1} \right] - \left(\mathbb{E} \left[e^{-\lambda_1(\zeta_n - y)} X_1 e^{\theta e^{-\lambda_1(\zeta_n - y)} X_1} \right] \right)^2}{\left(\mathbb{E} \left[e^{\theta e^{-\lambda_1(\zeta_n - y)} X_1} \right] \right)^2} \\
&> 0, \text{ and}
\end{aligned}$$

$$\begin{aligned}
\Lambda''_n(\theta) &= \frac{\int_0^{\zeta_n - y} e^{\lambda_0 t} \frac{e^{\lambda_1 t} e^{-3\lambda_1(\zeta_n - y)} e^{-\theta e^{-\lambda_1(\zeta_n - y)}} \left(e^{-\theta e^{-\lambda_1(\zeta_n - y)}} - e^{\lambda_1 t} e^{-\lambda_1(\zeta_n - y)} + 1 \right)}{\left(e^{\lambda_1 t} e^{-\lambda_1(\zeta_n - y)} - 1 + e^{-\theta e^{-\lambda_1(\zeta_n - y)}} \right)^3} dt}{\int_0^{\zeta_n - y} e^{\lambda_0 t} \frac{e^{\lambda_1 t} e^{-\lambda_1(\zeta_n - y)}}{e^{\lambda_1 t} e^{-\lambda_1(\zeta_n - y)} - 1 + e^{-\theta e^{-\lambda_1(\zeta_n - y)}}} dt} - (\Lambda'_n(\theta))^2 \\
&\rightarrow \frac{\int_0^\infty e^{\lambda_0 t} \frac{2e^{-2\lambda_1 t}}{(1 - \theta e^{-\lambda_1 t})^3} dt \int_0^\infty e^{\lambda_0 t} \frac{1}{1 - \theta e^{-\lambda_1 t}} dt - \left(\int_0^\infty e^{\lambda_0 t} \frac{e^{-\lambda_1 t}}{(1 - \theta e^{-\lambda_1 t})^2} dt \right)^2}{\left(\int_0^\infty e^{\lambda_0 t} \frac{1}{1 - \theta e^{-\lambda_1 t}} dt \right)^2} \text{ as } n \rightarrow \infty.
\end{aligned}$$

We denote the limit by $\Lambda''(\theta)$. Taking the third derivative, we have

$$\Lambda'''_n(\theta) = \frac{f_4(\theta) f_1^2(\theta) - 3f_3(\theta) f_2(\theta) f_1(\theta) + 2f_2^3(\theta)}{f_1^3(\theta)},$$

where

$$f_1(\theta) = \int_0^{\zeta_n - y} e^{\lambda_0 t} \frac{e^{\lambda_1 t} e^{-\lambda_1(\zeta_n - y)}}{e^{\lambda_1 t} e^{-\lambda_1(\zeta_n - y)} - 1 + e^{-\theta} e^{-\lambda_1(\zeta_n - y)}} dt, \quad (44)$$

$$f_2(\theta) = \int_0^{\zeta_n - y} e^{\lambda_0 t} \frac{e^{\lambda_1 t} e^{-2\lambda_1(\zeta_n - y)} e^{-\theta} e^{-\lambda_1(\zeta_n - y)}}{(e^{\lambda_1 t} e^{-\lambda_1(\zeta_n - y)} - 1 + e^{-\theta} e^{-\lambda_1(\zeta_n - y)})^2} dt, \quad (45)$$

$$f_3(\theta) = \int_0^{\zeta_n - y} e^{\lambda_0 t} \frac{e^{\lambda_1 t} e^{-3\lambda_1(\zeta_n - y)} e^{-\theta} e^{-\lambda_1(\zeta_n - y)} (e^{-\theta} e^{-\lambda_1(\zeta_n - y)} - e^{\lambda_1 t} e^{-\lambda_1(\zeta_n - y)} + 1)}{(e^{\lambda_1 t} e^{-\lambda_1(\zeta_n - y)} - 1 + e^{-\theta} e^{-\lambda_1(\zeta_n - y)})^3} dt, \quad (46)$$

$$f_4(\theta) = \int_0^{\zeta_n - y} e^{\lambda_0 t} \frac{e^{\lambda_1 t} e^{-4\lambda_1(\zeta_n - y)} e^{-\theta} e^{-\lambda_1(\zeta_n - y)} g_4(\theta)}{(e^{\lambda_1 t} e^{-\lambda_1(\zeta_n - y)} - 1 + e^{-\theta} e^{-\lambda_1(\zeta_n - y)})^4} dt, \text{ and} \quad (47)$$

$$g_4(\theta) = 4e^{-\theta} e^{-\lambda_1(\zeta_n - y)} - 4e^{\lambda_1 t} e^{-\theta} e^{-\lambda_1(\zeta_n - y)} e^{-\lambda_1(\zeta_n - y)} + e^{-2\theta} e^{-\lambda_1(\zeta_n - y)} \\ - 2e^{\lambda_1 t} e^{-\lambda_1(\zeta_n - y)} + e^{2\lambda_1 t} e^{-2\lambda_1(\zeta_n - y)} + 1.$$

By observing that $f_1(\theta)$ through $f_4(\theta)$ have well defined positive limits as n approaches infinity, for any $0 < \epsilon < 1$, we can find an $M_\epsilon > 0$ such that for sufficiently large n ,

$$|\Lambda_n'''(\theta)| \leq M_\epsilon \quad (48)$$

for $\theta \in (0, 1 - \epsilon)$. We then show the following lemma.

Lemma 10 *For sufficiently large n and sufficiently small ϵ_1 and ϵ_2 , we can find a unique solution $\eta_{k,n} \in (0, 1)$ to*

$$\Lambda_n'(\eta) = Q_{k,n}.$$

Proof See Section 8.1.2. □

We then show that θ_y^* is the solution to the equation in Lemma 10 when we take limits on both sides.

Lemma 11 $\Lambda'(\theta_y^*) = Q^*$.

Proof See Section 8.1.3. □

By Lemma 11 and the previous calculation of derivatives of $\Lambda_n(\theta)$, we can obtain very tight bounds for $\eta_{k,n}$, $Q_{k,n}$, $\Lambda_n(\eta_{k,n})$, $\Lambda_n'(\eta_{k,n})$ and $\Lambda_n''(\eta_{k,n})$ for sufficiently large n , where $\eta_{k,n}$ is the solution to $\Lambda_n'(\eta) = Q_{k,n}$.

Lemma 12 *Define*

$$Q_{k,n}[\epsilon_1] = \left(\inf_{k \in O_{n,y,\epsilon_1}} Q_{k,n}, \sup_{k \in O_{n,y,\epsilon_1}} Q_{k,n} \right),$$

and define $\eta_{k,n}[\epsilon_1]$, $\Lambda_n(\eta_{k,n})[\epsilon_1]$, $\Lambda'_n(\eta_{k,n})[\epsilon_1]$, $\Lambda''_n(\eta_{k,n})[\epsilon_1]$ in the same way. For any $\epsilon > 0$, there exists $\delta_\epsilon > 0$ and $N_\epsilon > 0$ such that when $\epsilon_1 < \delta_\epsilon$ and $n > N_\epsilon$,

$$\begin{aligned} Q_{k,n}[\epsilon_1] &\subset (Q^* - \epsilon, Q^* + \epsilon) \\ \eta_{k,n}[\epsilon_1] &\subset (\theta_y^* - \epsilon, \theta_y^* + \epsilon) \\ \Lambda_n(\eta_{k,n})[\epsilon_1] &\subset (\Lambda(\theta_y^*) - \epsilon, \Lambda(\theta_y^*) + \epsilon) \\ \Lambda'_n(\eta_{k,n})[\epsilon_1] &\subset (\Lambda'(\theta_y^*) - \epsilon, \Lambda'(\theta_y^*) + \epsilon) \\ \Lambda''_n(\eta_{k,n})[\epsilon_1] &\subset (\Lambda''(\theta_y^*) - \epsilon, \Lambda''(\theta_y^*) + \epsilon). \end{aligned}$$

Proof See Section 8.1.4. □

We then show the following proposition which gives a bound on (40).

Proposition 6 *Let $\nu_{n,y}$ be the law of $e^{-\lambda_1(\zeta_n - y)} X_1$, and consider the probability measure $\tilde{\nu}_{n,y,k}$ defined by $d\tilde{\nu}_{n,y,k}/d\nu_{n,y}(x) = e^{\eta_{k,n}x - \Lambda_n(\eta_{k,n})}$. For any $\epsilon > 0$, there exists $N_\epsilon > 0$ and $\delta_\epsilon > 0$ such that when $n > N_\epsilon$, $\epsilon_1 < \delta_\epsilon$, and $k \in O_{n,y,\epsilon_1}$,*

$$\begin{aligned} &\mathbb{E} \left[\left(e^{-\lambda_1(\zeta_n - y)} X_1 \right)^2 \middle| \frac{\sum_{i=1}^k e^{-\lambda_1(\zeta_n - y)} X_i}{k} \geq Q_{k,n} \right] \\ &\in \left(\int_0^\infty x^2 d\tilde{\nu}_{n,y,k}(x) - \epsilon, \int_0^\infty x^2 d\tilde{\nu}_{n,y,k}(x) + \epsilon \right). \end{aligned}$$

Proof See Section 8.1.5. □

Notice that (see page 111 of Dembo and Zeitouni (2009))

$$\int_0^\infty x^2 d\tilde{\nu}_{n,y,k}(x) = \Lambda''_n(\eta_{k,n}) + Q_{k,n}^2.$$

By Lemma 12, we know that for any $\epsilon > 0$, the following result holds for sufficiently small ϵ_1 , sufficiently large n and $k \in O_{n,y,\epsilon_1}$:

$$\int_0^\infty x^2 d\tilde{\nu}_{n,y,k} \in (\Lambda''(\theta_y^*) + Q^{*2} - \epsilon, \Lambda''(\theta_y^*) + Q^{*2} + \epsilon).$$

Hence

$$\begin{aligned}
& \lim_{n \rightarrow \infty} n^{1-\alpha} \mathbb{E}_{A_{n,y}} [R_{n,y}] \\
&= \lim_{n \rightarrow \infty} n^{1-\alpha} \mathbb{E}_{E_{n,y,\epsilon_1}} [R_{n,y}] \\
&= \lim_{n \rightarrow \infty} n^{1-\alpha} \sum_{k \in O_{n,y,\epsilon_1}} \mathbb{E} [R_{n,y} | \gamma_n < \zeta_n - y, I_n(\zeta_n - y) = k] \frac{\mathbb{P}(\gamma_n < \zeta_n - y, I_n(\zeta_n - y) = k)}{\mathbb{P}(E_{n,y,\epsilon_1})} \\
&= \lim_{n \rightarrow \infty} n^{1-\alpha} \sum_{k \in O_{n,y,\epsilon_1}} \mathbb{E} [\tilde{R}_{n,y} | \gamma_n < \zeta_n - y, I_n(\zeta_n - y) = k] \frac{\mathbb{P}(\gamma_n < \zeta_n - y, I_n(\zeta_n - y) = k)}{\mathbb{P}(E_{n,y,\epsilon_1})} \\
&= \lim_{n \rightarrow \infty} n^{1-\alpha} \sum_{k \in O_{n,y,\epsilon_1}} \frac{k e^{2\lambda_1(\zeta_n - y)}}{n^2} \mathbb{E} \left[\left(e^{-\lambda_1(\zeta_n - y)} X_i \right)^2 \middle| \sum_{i=1}^k e^{-\lambda_1(\zeta_n - y)} X_i \geq e^{-\lambda_1(\zeta_n - y)} n \right] \\
&\quad \frac{\mathbb{P}(\gamma_n < \zeta_n - y, I_n(\zeta_n - y) = k)}{\mathbb{P}(E_{n,y,\epsilon_1})},
\end{aligned}$$

where the last equality follows from (39). By Proposition 6, the definition of O_{n,y,ϵ_1} and $\tilde{O}_{y,\epsilon_1,\epsilon_2}$, and the fact that ϵ_1 and ϵ_2 can be arbitrarily small, we have

$$\begin{aligned}
& \lim_{n \rightarrow \infty} n^{1-\alpha} \mathbb{E}_{A_{n,y}} [R_{n,y}] \\
&= \lim_{n \rightarrow \infty} n^{1-\alpha} \frac{\mu n^{1-\alpha} \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta_y^*} e^{\lambda_0 s} ds}{n^2} e^{2\lambda_1(\zeta_n - y)} \left(\Lambda''(\theta_y^*) + Q^{*2} \right) \\
&= \lim_{n \rightarrow \infty} n^{1-\alpha} \frac{\mu n^{1-\alpha} \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta_y^*} e^{\lambda_0 s} ds}{n^2} e^{2\lambda_1(\zeta_n - y)} \frac{2 \int_0^{\zeta_n - y} \frac{e^{-(2\lambda_1 - \lambda_0)s}}{(1 - \theta_y^* e^{-\lambda_1 s})^3} ds}{\int_0^{\zeta_n - y} \frac{e^{\lambda_0 s}}{1 - \theta_y^* e^{-\lambda_1 s}} ds} \\
&= \frac{2(\lambda_1 - \lambda_0)^2}{\mu} e^{-2\lambda_1 y} \int_0^\infty \frac{e^{-(2\lambda_1 - \lambda_0)s}}{(1 - \theta_y^* e^{-\lambda_1 s})^3} ds.
\end{aligned}$$

□

8.1.1 Proof of Lemma 9

Proof Let $z = -\lambda_0 \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta_y^*} e^{\lambda_0 s} ds$, which is the optimal value obtained in Lemma 8. We first notice that

$$A_{n,y} \setminus E_{n,y,\epsilon_1} = \{\gamma_n < \zeta_n - y, I_n(\zeta_n - y) \in O_1\} \cup \{\gamma_n < \zeta_n - y, I_n(\zeta_n - y) \in O_2\},$$

where

$$O_1 = \left(0, -(1 - \epsilon_1) z \frac{\mu}{\lambda_0} n^{1-\alpha} \right), \text{ and } O_2 = \left(-(1 + \epsilon_1) z \frac{\mu}{\lambda_0} n^{1-\alpha}, \infty \right).$$

We consider O_2 first. It is easy to show that the probability of $\{\gamma_n < \zeta_n - y, I_n(\zeta_n - y) \geq -e^{\lambda_1 y} \frac{\mu}{\lambda_0} n^{1-\alpha}\}$ decays exponentially faster than that of $A_{n,y}$. Hence,

we focus on the interval $((1 + \epsilon_1)z, e^{\lambda_1 y})$, and divide it into M sub-intervals with equal length. Let $\delta_M = \frac{e^{\lambda_1 y} - (1 + \epsilon_1)z}{M}$ be the length of each sub-interval. Then for $i \in \{1, 2, \dots, M\}$, we consider the sub-interval

$$O_{i,M} = ((1 + \epsilon_1)z + (i - 1)\delta_M, (1 + \epsilon_1)z + i\delta_M).$$

By stochastic dominance and the Gartner-Ellis Theorem (Hollander 2000), we can show that

$$\begin{aligned} & \limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P} \left(\gamma_n \leq \zeta_n - y, -\frac{\lambda_0}{\mu} n^{\alpha-1} I_n(\zeta_n - y) \in O_{i,M} \right) \\ & \leq \limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \left(\mathbb{P} \left(\gamma_n \leq \zeta_n - y \mid I_n(\zeta_n - y) = \lfloor -\frac{\mu}{\lambda_0} n^{1-\alpha} ((1 + \epsilon_1)z + i\delta_M) \rfloor \right) \mathbb{P} \right. \\ & \quad \left. \left(-\frac{\lambda_0}{\mu} n^{\alpha-1} I_n(\zeta_n - y) \in O_{i,M} \right) \right) \\ & = - \sup_{\theta \in (0,1)} \left[\frac{\theta \mu e^{\lambda_1 y}}{\lambda_1 - \lambda_0} - \lfloor (1 + \epsilon_1)z + i\delta_M \rfloor \frac{\mu}{\lambda_0} \log \left(-\lambda_0 \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta} e^{\lambda_0 s} ds \right) \right] \\ & \quad + \frac{\mu}{\lambda_0} (\lfloor (1 + \epsilon_1)z + (i - 1)\delta_M \rfloor \log (\lfloor (1 + \epsilon_1)z + (i - 1)\delta_M \rfloor) - \lfloor (1 + \epsilon_1)z + (i - 1)\delta_M \rfloor + 1). \end{aligned}$$

Let

$$\begin{aligned} f_y(x) &= \sup_{\theta \in (0,1)} \left[\frac{\theta \mu e^{\lambda_1 y}}{\lambda_1 - \lambda_0} + x \frac{\mu}{\lambda_0} \log \left(-\lambda_0 \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta} e^{\lambda_0 s} ds \right) \right] \\ & \quad - \frac{\mu}{\lambda_0} (x \log(x) - x + 1). \end{aligned}$$

From Hanagal et al. (2022) and the Envelope Theorem, we know that $f_y(x)$ is strictly convex in x , and

$$z = \operatorname{argmin}_{x \in (0, e^{\lambda_1 y}]} f_y(x).$$

Moreover, θ_y^* is the corresponding optimal solution to the inner optimization problem. By some algebra we can obtain that $f_y(z) = L(y)$. Hence, we conclude that

$$\begin{aligned} L(y) &= \sup_{\theta \in (0,1)} \left[\frac{\theta \mu e^{\lambda_1 y}}{\lambda_1 - \lambda_0} - \lfloor (1 + \epsilon_1)z + i\delta_M \rfloor \frac{\mu}{\lambda_0} \log \left(-\lambda_0 \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta} e^{\lambda_0 s} ds \right) \right] \\ & \quad + \frac{\mu}{\lambda_0} (\lfloor (1 + \epsilon_1)z + i\delta_M \rfloor \log (\lfloor (1 + \epsilon_1)z + i\delta_M \rfloor) - \lfloor (1 + \epsilon_1)z + i\delta_M \rfloor + 1) \end{aligned}$$

is strictly less than 0 for any i and M . Therefore, there exists $\epsilon_2 > 0$ and an integer $M_{\epsilon_2} > 0$ such that for all $i \in \{1, 2, \dots, M_{\epsilon_2}\}$,

$$\begin{aligned} L(y) &= \sup_{\theta \in (0,1)} \left[\frac{\theta \mu e^{\lambda_1 y}}{\lambda_1 - \lambda_0} + \lfloor (1 + \epsilon_1)z + i\delta_{M_{\epsilon_2}} \rfloor \frac{\mu}{\lambda_0} \log \left(-\lambda_0 \int_0^\infty \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \theta} e^{\lambda_0 s} ds \right) \right] \\ & \quad + \frac{\mu}{\lambda_0} (\lfloor (1 + \epsilon_1)z + (i - 1)\delta_{M_{\epsilon_2}} \rfloor \log (\lfloor (1 + \epsilon_1)z + (i - 1)\delta_{M_{\epsilon_2}} \rfloor) \end{aligned}$$

$$\begin{aligned}
& -\lfloor (1 + \epsilon_1)z + (i - 1)\delta_{M_{\epsilon_2}} \rfloor + 1) \\
& < -\epsilon_2.
\end{aligned}$$

Hence, we conclude that

$$\begin{aligned}
& \limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \log \mathbb{P} \left(\gamma_n \leq \zeta_n - y, -\frac{\lambda_0}{\mu} n^{\alpha-1} I_n(\zeta_n - y) \in O_{i,M} \right) \\
& < -L(y).
\end{aligned}$$

We can get a similar result for the interval $(0, (1 - \epsilon_1)z)$ and the desired result follows. $\square$

8.1.2 Proof of Lemma 10

Proof Since $\Lambda''_n(\theta) > 0$, $\Lambda'_n(\theta)$ increases in θ . By (41), (42), (43), and the fact that $q^* > 1$, we can obtain that $\Lambda'_n(0) = \frac{\int_0^{\zeta_n-y} e^{-(\lambda_1-\lambda_0)t} dt}{\int_0^{\zeta_n-y} e^{\lambda_0 t} dt} < Q_{k,n}$ when ϵ_1 and ϵ_2 are sufficiently small, and n is sufficiently large. Since

$$\begin{aligned}
& \lim_{\theta \rightarrow 1} \Lambda'(\theta) = \infty, \text{ and} \\
& \lim_{n \rightarrow \infty} \Lambda'_n(\theta) = \Lambda'(\theta),
\end{aligned}$$

we can find a sufficiently small $\epsilon > 0$ such that for sufficiently large n ,

$$\Lambda'_n(1 - \epsilon) > Q_{k,n},$$

which leads to the desired result. $\square$

8.1.3 Proof of Lemma 11

Proof By the definition of $\Lambda'(\theta)$ and Q^* , and (4), it is easy to verify that $\Lambda'(\theta_y^*) = Q^*$. $\square$

8.1.4 Proof of Lemma 12

Proof The result for $Q_{k,n}$ follows immediately from the definition for $Q_{k,n}$ and Q^* . We know that $\eta_{k,n} = \Lambda'^{-1}_n(Q_{k,n})$. Since $\Lambda'_n(\theta)$ is a continuous and strictly increasing function, its inverse is also continuous and strictly increasing. The result for $\eta_{k,n}$ then follows immediately from the result for $Q_{k,n}$. The rest of the results for $\Lambda_n(\eta_{k,n})$, and $\Lambda'_n(\eta_{k,n})$ follow immediately from their monotonicity. The result of $\Lambda''_n(\eta_{k,n})$ follows from (48). $\square$

8.1.5 Proof of Proposition 6

Proof We follow the main idea of the proof of Theorem 3.7.4 (Bahadur and Rao) of Dembo and Zeitouni (2009). Define

$$Y_i = \frac{e^{-\lambda_1(\zeta_n - y)} X_i - Q_{k,n}}{\sqrt{\Lambda''_n(\eta_{k,n})}}.$$

It is easy to verify that $\mathbb{E}_{\tilde{\nu}_{n,y,k}}[Y_i] = 0$, $\mathbb{E}_{\tilde{\nu}_{n,y,k}}[Y_i^2] = 1$, and $\mathbb{E}_{\tilde{\nu}_{n,y,k}}[Y_i^3]$ exists. We let

$$\begin{aligned}\alpha_{3,n,y,q_{k,n}} &= \mathbb{E}_{\tilde{\nu}_{n,y,k}}[Y_i^3], \\ \beta_{3,n,y,q_{k,n}} &= \mathbb{E}_{\tilde{\nu}_{n,y,k}}[|Y_i|^3].\end{aligned}$$

By Lemma 11, Lemma 12 and the previous calculation of derivatives of $\Lambda_n(\theta)$, we can obtain useful bounds for $\alpha_{3,n,y,q_{k,n}}$ and $\beta_{3,n,y,q_{k,n}}$. Let the probability measure $\tilde{\nu}_{n,y}$ be defined by $d\tilde{\nu}_{n,y}/d\nu_{n,y}(x) = e^{\theta_y^* x - \Lambda(\eta_y^*)}$.

Lemma 13 *Define*

$$\alpha_{3,n,y,q_{k,n}}[\epsilon_1] = \left(\inf_{k \in O_{n,y,\epsilon_1}} \alpha_{3,n,y,q_{k,n}}, \sup_{k \in O_{n,y,\epsilon_1}} \alpha_{3,n,y,q_{k,n}} \right).$$

For any $\epsilon > 0$, there exists $\delta_\epsilon > 0$ and $N_\epsilon > 0$ such that when $\epsilon_1 < \delta_\epsilon$ and $n > N_\epsilon$,

$$\alpha_{3,n,y,q_{k,n}}[\epsilon_1] \subset (\alpha_3^* - \epsilon, \alpha_3^* + \epsilon),$$

where

$$\alpha_3^* = \lim_{n \rightarrow \infty} \mathbb{E}_{\tilde{\nu}_{n,y}} \left[\left(\frac{e^{-\lambda_1(\zeta_n - y)} X_1 - Q^*}{\sqrt{\Lambda''_n(\theta_y^*)}} \right)^3 \right].$$

Moreover,

$$\limsup_{\epsilon_1 \rightarrow 0} \limsup_{n \rightarrow \infty} \sup_{k \in O_{n,y,\epsilon_1}} \beta_{3,n,y,q_{k,n}} \leq \lim_{n \rightarrow \infty} \mathbb{E}_{\tilde{\nu}_{n,y}} \left[\left(\frac{e^{-\lambda_1(\zeta_n - y)} X_1 - Q^*}{\sqrt{\Lambda''_n(\theta_y^*)}} \right)^6 \right], \text{ and}$$

$$\beta_{3,n,y,q_{k,n}} \geq 1.$$

Proof See Section 8.1.6. $\square$

For ease of exposition, we omit the subscript n , y , and $q_{k,n}$, and just refer to them as α_3 and β_3 . Let $F_{k,n}$ be the distribution function of

$$k^{-\frac{1}{2}} \sum_{i=1}^k Y_i$$

when $e^{-\lambda_1(\zeta_n - y)} X_i$ are i.i.d. with marginal law $\tilde{v}_{n,y,k}$. Let $\phi(x)$ be the pdf and $\Phi(x)$ be the cdf of a standard normal distribution. We know that

$$\phi'(x) = -\frac{x}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}.$$

A key step in our proof is to show a modified version of the Berry-Esseen expansion for lattice distributions (Theorem 3 in Chapter IV of Esseen (1945) and Theorem 1 in §43 of Gnedenko and Kolmogorov (1954)). Compared to the original version, we need to consider a parameter-dependent span (notice that Y_i has a span of $\frac{e^{-\lambda_1(\zeta_n - y)}}{\sqrt{\Lambda_n''(\eta_{k,n})}}$) instead of a fixed one. Moreover, we need the convergence to be uniform for $k \in O_{n,y,\epsilon_1}$ as n goes to infinity. We first introduce some notation. Let

$$\begin{aligned} h_{k,n} &= \frac{e^{-\lambda_1(\zeta_n - y)}}{\sqrt{\Lambda_n''(\eta_{k,n})}}, \\ \tau_{k,n} &= \frac{2\pi}{h_{k,n}}, \\ x_{k,n} &= -\frac{Q_{k,n}}{\sqrt{\Lambda_n''(\eta_{k,n})}}, \text{ and} \\ S(x) &= [x] - x + \frac{1}{2}. \end{aligned}$$

where $[x]$ gives the integral part of x . Notice that $h_{k,n}$ is the span of Y_i and $\tau_{k,n}$ is the period of its. Let

$$S_{k,n}(x) = h_{k,n} S\left(\frac{x\sqrt{k} - x_{k,n}k}{h_{k,n}}\right), \text{ and}$$

$$D_{k,n}(x) = \phi(x) \frac{S_{k,n}(x)}{\sqrt{k}}.$$

Lemma 14 For any $\epsilon > 0$, there exists $N_\epsilon > 0$ and $\delta_\epsilon > 0$ such that when $n > N_\epsilon$ and $\epsilon_1 < \delta_\epsilon$,

$$\sup_{k \in O_{n,y,\epsilon_1}} \left\{ \sqrt{k} \sup_x \left| F_{k,n}(x) - \Phi(x) - \frac{\alpha_3}{6\sqrt{k}} (1-x^2) \phi(x) - D_{k,n}(x) \right| \right\} < \epsilon.$$

Proof See Section 8.1.7. □

Let

$$\begin{aligned} \psi_{k,n} &= \eta_{k,n} \sqrt{k \Lambda_n''(\eta_{k,n})}, \text{ and} \\ J_{n,y,k} &= \eta_{k,n} \sqrt{\Lambda_n''(\eta_{k,n})} 2\pi k e^{k \Lambda_n^*(Q_{k,n})}, \end{aligned}$$

where $\Lambda_n^*(Q_{k,n}) = \eta_{k,n} Q_{k,n} - \Lambda_n(\eta_{k,n})$. Let

$$\begin{aligned} C_{n,y,k} &= \sqrt{2\pi} \int_0^\infty \psi_{k,n} e^{-t} \left(\Phi\left(\frac{t}{\psi_{k,n}}\right) + \frac{\alpha_3}{6\sqrt{k}} \left(1 - \left(\frac{t}{\psi_{k,n}}\right)^2\right) \phi\left(\frac{t}{\psi_{k,n}}\right) + \phi\left(\frac{t}{\psi_{k,n}}\right) \frac{S_{k,n}\left(\frac{t}{\psi_{k,n}}\right)}{\sqrt{k}} \right) dt \\ &\quad - \sqrt{2\pi} \int_0^\infty \psi_{k,n} e^{-t} \left(\Phi(0) + \frac{\alpha_3}{6\sqrt{k}} \phi(0) + \phi(0) \frac{S_{k,n}(0)}{\sqrt{k}} \right) dt. \end{aligned}$$

We then show the following lemma which provides a good approximation for

$$\mathbb{P} \left(\sum_{i=1}^k e^{-\lambda_1(\zeta_n - y)} X_i \geq k Q_{k,n} \right).$$

Lemma 15 For any $\epsilon > 0$, there exists $N_\epsilon > 0$ and $\delta_\epsilon > 0$ such that when $n > N_\epsilon$ and $\epsilon_1 < \delta_\epsilon$,

$$\sup_{k \in O_{n,y,\epsilon_1}} \left| \mathbb{P} \left(\sum_{i=1}^k e^{-\lambda_1(\zeta_n - y)} X_i \geq k Q_{k,n} \right) J_{n,y,k} - C_{n,y,k} \right| \leq \epsilon.$$

Proof See Section 8.1.8. □

We then analyze $C_{n,y,k}$. By a Taylor expansion of $\Phi\left(\frac{t}{\psi_{k,n}}\right)$, we have

$$\begin{aligned} &\Phi\left(\frac{t}{\psi_{k,n}}\right) + \frac{\alpha_3}{6\sqrt{k}} \left(1 - \left(\frac{t}{\psi_{k,n}}\right)^2\right) \phi\left(\frac{t}{\psi_{k,n}}\right) + \phi\left(\frac{t}{\psi_{k,n}}\right) \frac{S_{k,n}\left(\frac{t}{\psi_{k,n}}\right)}{\sqrt{k}} \\ &\quad - \Phi(0) - \frac{\alpha_3}{6\sqrt{k}} \phi(0) - \phi(0) \frac{S_{k,n}(0)}{\sqrt{k}} \\ &= \frac{\alpha_3}{6\sqrt{k}} \left(1 - \left(\frac{t}{\psi_{k,n}}\right)^2\right) \phi\left(\frac{t}{\psi_{k,n}}\right) - \frac{z\phi(z)}{2} \left(\frac{t}{\psi_{k,n}}\right)^2 + \phi(0) \left(\frac{t}{\psi_{k,n}}\right) \\ &\quad - \frac{\alpha_3}{6\sqrt{k}} \phi(0) + \phi\left(\frac{t}{\psi_{k,n}}\right) \frac{S_{k,n}\left(\frac{t}{\psi_{k,n}}\right)}{\sqrt{k}} - \phi(0) \frac{S_{k,n}(0)}{\sqrt{k}}, \end{aligned}$$

where z is some number between 0 and $\frac{t}{\psi_{k,n}}$. Dividing the integral in $C_{n,y,k}$ into two parts, we have

$$\begin{aligned}
 C_{n,y,k} &= \sqrt{2\pi} \int_0^\infty \psi_{k,n} e^{-t} \left(\frac{\alpha_3}{6\sqrt{k}} \left(1 - \left(\frac{t}{\psi_{k,n}} \right)^2 \right) \phi \left(\frac{t}{\psi_{k,n}} \right) + \phi(0) \left(\frac{t}{\psi_{k,n}} \right) \right. \\
 &\quad \left. + \phi \left(\frac{t}{\psi_{k,n}} \right) \frac{S_{k,n} \left(\frac{t}{\psi_{k,n}} \right)}{\sqrt{k}} \right) dt \\
 &\quad - \sqrt{2\pi} \int_0^\infty \psi_{k,n} e^{-t} \left(\frac{z\phi(z)}{2} \left(\frac{t}{\psi_{k,n}} \right)^2 + \frac{\alpha_3}{6\sqrt{k}} \phi(0) + \phi(0) \frac{S_{k,n}(0)}{\sqrt{k}} \right) dt \\
 &= \sqrt{2\pi} \int_0^T \psi_{k,n} e^{-t} \left(\frac{\alpha_3}{6\sqrt{k}} \left(1 - \left(\frac{t}{\psi_{k,n}} \right)^2 \right) \phi \left(\frac{t}{\psi_{k,n}} \right) \right. \\
 &\quad \left. + \phi(0) \left(\frac{t}{\psi_{k,n}} \right) + \phi \left(\frac{t}{\psi_{k,n}} \right) \frac{S_{k,n} \left(\frac{t}{\psi_{k,n}} \right)}{\sqrt{k}} \right) dt \\
 &\quad - \sqrt{2\pi} \int_0^T \psi_{k,n} e^{-t} \left(\frac{z\phi(z)}{2} \left(\frac{t}{\psi_{k,n}} \right)^2 + \frac{\alpha_3}{6\sqrt{k}} \phi(0) + \phi(0) \frac{S_{k,n}(0)}{\sqrt{k}} \right) dt \\
 &\quad + \sqrt{2\pi} \int_T^\infty \psi_{k,n} e^{-t} \left(\frac{\alpha_3}{6\sqrt{k}} \left(1 - \left(\frac{t}{\psi_{k,n}} \right)^2 \right) \phi \left(\frac{t}{\psi_{k,n}} \right) \right. \\
 &\quad \left. + \phi(0) \left(\frac{t}{\psi_{k,n}} \right) + \phi \left(\frac{t}{\psi_{k,n}} \right) \frac{S_{k,n} \left(\frac{t}{\psi_{k,n}} \right)}{\sqrt{k}} \right) dt \\
 &\quad - \sqrt{2\pi} \int_T^\infty \psi_{k,n} e^{-t} \left(\frac{z\phi(z)}{2} \left(\frac{t}{\psi_{k,n}} \right)^2 + \frac{\alpha_3}{6\sqrt{k}} \phi(0) + \phi(0) \frac{S_{k,n}(0)}{\sqrt{k}} \right) dt.
 \end{aligned}$$

where T can be arbitrarily large but fixed. By the definition of $\phi(x)$, the second part goes to 0 as T goes to infinity (note that the convergence is uniform for $k \in O_{n,y,\epsilon_1}$ for sufficiently large n and sufficiently small ϵ_1 by Lemma 12 and Lemma 13). We then analyze the first part. By dominated convergence theorem and Lemma 12,

$$\begin{aligned}
 &\lim_{n \rightarrow \infty} \sqrt{2\pi} \int_0^T \psi_{k,n} e^{-t} \left(\frac{\alpha_3}{6\sqrt{k}} \left(1 - \left(\frac{t}{\psi_{k,n}} \right)^2 \right) \phi \left(\frac{t}{\psi_{k,n}} \right) + \phi(0) \left(\frac{t}{\psi_{k,n}} \right) \right. \\
 &\quad \left. + \phi \left(\frac{t}{\psi_{k,n}} \right) \frac{S_{k,n} \left(\frac{t}{\psi_{k,n}} \right)}{\sqrt{k}} \right) dt \\
 &\quad - \lim_{n \rightarrow \infty} \sqrt{2\pi} \int_0^T \psi_{k,n} e^{-t} \left(\frac{z\phi(z)}{2} \left(\frac{t}{\psi_{k,n}} \right)^2 + \frac{\alpha_3}{6\sqrt{k}} \phi(0) + \phi(0) \frac{S_{k,n}(0)}{\sqrt{k}} \right) dt \\
 &= \sqrt{2\pi} \phi(0) \int_0^T t e^{-t} dt \\
 &= -(T+1)e^{-T} + 1,
 \end{aligned}$$

where the convergence is uniform for $k \in O_{n,y,\epsilon_1}$. Hence, when ϵ_1 is sufficiently small,

$$\lim_{n \rightarrow \infty} \sup_{k \in O_{n,y,\epsilon_1}} |C_{n,y,k} - 1| = 0. \quad (49)$$

Following the proof of Theorem 3.7.4 of Dembo and Zeitouni (2009) we have

$$\begin{aligned} & \mathbb{E} \left[\left(e^{-\lambda_1(\zeta_n - y)} X_i \right)^2 \middle| \sum_{i=1}^k X_i \geq n \right] \\ &= 1/\mathbb{P} \left(\sum_{i=1}^k X_i \geq n \right) \times \int_0^\infty x^2 e^{-\eta_{k,n}x + \Lambda_n(\eta_{k,n})} e^{-(k-1)\Lambda_n^*(Q_{k,n})} \\ & \quad \int_{\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}}}^\infty e^{-\sqrt{\frac{k-1}{k}}\psi_{k,n}z} dF_{k-1,n}(z) d\tilde{v}_{n,y,k}(x), \end{aligned}$$

where $\Lambda_n^*(Q_{k,n}) = \eta_{k,n}Q_{k,n} - \Lambda_n(\eta_{k,n})$, and the lower bound of the inner integral comes from the following calculation:

$$\begin{aligned} (k-1)^{-1/2} \sum_{i=1}^{k-1} Y_i &= (k-1)^{-1/2} \sum_{i=1}^{k-1} \left(e^{-\lambda_1(\zeta_n - y)} X_i - Q_{k,n} \right) / \sqrt{\Lambda_n''(\eta_{k,n})} \\ &\geq \frac{1}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}} (Q_{k,n}k - x) - \frac{\sqrt{k-1}Q_{k,n}}{\sqrt{\Lambda_n''(\eta_{k,n})}} \\ &= \frac{Q_{k,n} - x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}}. \end{aligned}$$

By Lemma 15 and 49, we know that for any $\epsilon_3 > 0$, the following holds for sufficiently small ϵ_1 and sufficiently large n :

$$\sup_{k \in O_{n,y,\epsilon_1}} \left| \mathbb{P} \left(\sum_{i=1}^k e^{-\lambda_1(\zeta_n - y)} X_i \geq kQ_{k,n} \right) J_{n,y,k} - 1 \right| < \epsilon_3.$$

Hence, we will work with

$$\begin{aligned} & \mathbb{E} \left[\left(e^{-\lambda_1(\zeta_n - y)} X_1 \right)^2 \middle| \sum_{i=1}^k X_i \geq n \right] J_{n,y,k} \mathbb{P} \left(\sum_{i=1}^k e^{-\lambda_1(\zeta_n - y)} X_i \geq kQ_{k,n} \right) \\ &= \eta_{k,n} \sqrt{2\pi k \Lambda_n''(\eta_{k,n})} e^{k\Lambda_n^*(Q_{k,n})} \int_0^\infty x^2 e^{-\eta_{k,n}x + \Lambda_n(\eta_{k,n})} e^{-(k-1)\Lambda_n^*(Q_{k,n})} \\ & \quad \int_{\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}}}^\infty e^{-\sqrt{\frac{k-1}{k}}\psi_{k,n}z} dF_{k-1,n}(z) d\tilde{v}_{n,y,k}(x) \end{aligned}$$

$$= \eta_{k,n} \sqrt{2\pi k \Lambda_n''(\eta_{k,n})} e^{\Lambda_n^*(Q_{k,n})} \int_0^\infty x^2 e^{-\eta_{k,n}x + \Lambda_n(\eta_{k,n})} \\ \int_{\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}}}^\infty e^{-\sqrt{\frac{k-1}{k}}\psi_{k,n}z} dF_{k-1,n}(z) d\tilde{v}_{n,y,k}(x),$$

where we use $\mathbb{P}\left(\sum_{i=1}^k e^{-\lambda_1(\zeta_n-y)} X_i \geq k Q_{k,n}\right) = \mathbb{P}\left(\sum_{i=1}^k X_i \geq n\right)$ in the first equality. Applying an integration by parts to the inner integral, we have

$$\begin{aligned} & \eta_{k,n} \sqrt{2\pi k \Lambda_n''(\eta_{k,n})} e^{\Lambda_n^*(Q_{k,n})} \int_0^\infty x^2 e^{-\eta_{k,n}x + \Lambda_n(\eta_{k,n})} \\ & \int_{\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}}}^\infty e^{-\sqrt{\frac{k-1}{k}}\psi_{k,n}z} dF_{k-1,n}(z) d\tilde{v}_{n,y,k}(x) \\ &= \eta_{k,n} \sqrt{2\pi k \Lambda_n''(\eta_{k,n})} e^{\Lambda_n^*(Q_{k,n})} \int_0^\infty x^2 e^{-\eta_{k,n}x + \Lambda_n(\eta_{k,n})} \\ & \times \int_{\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}}}^\infty \sqrt{\frac{k-1}{k}} \psi_{k,n} e^{-\sqrt{\frac{k-1}{k}}\psi_{k,n}z} \\ & \left[F_{k-1,n}(z) - F_{k-1,n}\left(\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}}\right) \right] dz d\tilde{v}_{n,y,k}(x). \quad (50) \end{aligned}$$

By a change of variable ($t = \sqrt{\frac{k-1}{k}}\psi_{k,n}z$),

$$\begin{aligned} (50) &= \frac{\eta_{k,n} \sqrt{2\pi k \Lambda_n''(\eta_{k,n})}}{\eta_{k,n} \sqrt{(k-1)\Lambda_n''(\eta_{k,n})}} e^{\Lambda_n^*(Q_{k,n})} \int_0^\infty x^2 e^{-\eta_{k,n}x + \Lambda_n(\eta_{k,n})} \\ & \times \int_{\eta_{k,n}(Q_{k,n}-x)}^\infty \sqrt{\frac{k-1}{k}} \psi_{k,n} e^{-t} \left[F_{k-1,n}\left(\frac{t}{\sqrt{\frac{k-1}{k}}\psi_{k,n}}\right) \right. \\ & \left. - F_{k-1,n}\left(\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}}\right) \right] dt d\tilde{v}_{n,y,k}(x) \\ &= \sqrt{2\pi \frac{k}{k-1}} e^{\Lambda_n^*(Q_{k,n})} \int_0^\infty x^2 e^{-\eta_{k,n}x + \Lambda_n(\eta_{k,n})} \\ & \times \int_{\eta_{k,n}(Q_{k,n}-x)}^\infty \sqrt{\frac{k-1}{k}} \psi_{k,n} e^{-t} \left[F_{k-1,n}\left(\frac{t}{\sqrt{\frac{k-1}{k}}\psi_{k,n}}\right) \right. \\ & \left. - F_{k-1,n}\left(\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}}\right) \right] dt d\tilde{v}_{n,y,k}(x). \quad (51) \end{aligned}$$

$$\begin{aligned} & \times \int_{\eta_{k,n}(Q_{k,n}-x)}^\infty \sqrt{\frac{k-1}{k}} \psi_{k,n} e^{-t} \left[F_{k-1,n}\left(\frac{t}{\sqrt{\frac{k-1}{k}}\psi_{k,n}}\right) \right. \\ & \left. - F_{k-1,n}\left(\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}}\right) \right] dt d\tilde{v}_{n,y,k}(x). \quad (52) \end{aligned}$$

Let

$$L_{1,n,y,k}(x) = e^{-\eta_{k,n}x} \int_{\eta_{k,n}(Q_{k,n}-x)}^{\infty} \sqrt{\frac{k-1}{k}} \psi_{k,n} e^{-t} \left[F_{k-1,n} \left(\frac{t}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} \right) - F_{k-1,n} \left(\frac{Q_{k,n}-x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right) \right] dt.$$

Let

$$\begin{aligned} L_{2,n,y,k}(x) &= e^{-\eta_{k,n}x} \int_{\eta_{k,n}(Q_{k,n}-x)}^{\infty} \sqrt{\frac{k-1}{k}} \psi_{k,n} e^{-t} \left(\Phi \left(\frac{t}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} \right) \right. \\ &\quad + \frac{\alpha_3}{6\sqrt{k-1}} \left(1 - \left(\frac{t}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} \right)^2 \right) \phi \left(\frac{t}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} \right) \\ &\quad + \phi \left(\frac{t}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} \right) \frac{S_{k-1,n} \left(\frac{t}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} \right)}{\sqrt{k-1}} - \Phi \left(\frac{Q_{k,n}-x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right) \\ &\quad - \frac{\alpha_3}{6\sqrt{k-1}} \left(1 - \left(\frac{Q_{k,n}-x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right)^2 \right) \phi \left(\frac{Q_{k,n}-x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right) \\ &\quad \left. - \phi \left(\frac{Q_{k,n}-x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right) \frac{S_{k-1,n} \left(\frac{Q_{k,n}-x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right)}{\sqrt{k-1}} \right) dt. \end{aligned}$$

By a change of variable ($z = t + \eta_{k,n}x$), we have

$$\begin{aligned} L_{1,n,y,k}(x) &= \int_{\eta_{k,n}Q_{k,n}}^{\infty} \sqrt{\frac{k-1}{k}} \psi_{k,n} e^{-z} \left[F_{k-1,n} \left(\frac{z - \eta_{k,n}x}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} \right) - F_{k-1,n} \left(\frac{Q_{k,n}-x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right) \right] dz, \end{aligned}$$

and

$$\begin{aligned}
 L_{2,n,y,k}(x) = & \int_{\eta_{k,n} Q_{k,n}}^{\infty} \sqrt{\frac{k-1}{k}} \psi_{k,n} e^{-z} \left(\Phi \left(\frac{z - \eta_{k,n} x}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} \right) \right. \\
 & + \frac{\alpha_3}{6\sqrt{k-1}} \left(1 - \left(\frac{z - \eta_{k,n} x}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} \right)^2 \right) \phi \left(\frac{z - \eta_{k,n} x}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} \right) \\
 & + \phi \left(\frac{z - \eta_{k,n} x}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} \right) \frac{S_{k-1,n} \left(\frac{z - \eta_{k,n} x}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} \right)}{\sqrt{k-1}} - \Phi \left(\frac{Q_{k,n} - x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right) \\
 & - \frac{\alpha_3}{6\sqrt{k-1}} \left(1 - \left(\frac{Q_{k,n} - x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right)^2 \right) \phi \left(\frac{Q_{k,n} - x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right) \\
 & \left. - \phi \left(\frac{Q_{k,n} - x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right) \frac{S_{k-1,n} \left(\frac{Q_{k,n} - x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right)}{\sqrt{k-1}} \right) dz.
 \end{aligned}$$

By Lemma 14, for any $\epsilon_4 > 0$ and any x , when ϵ_1 is sufficiently small and n is sufficiently large,

$$\sup_{k \in \mathcal{O}_{n,y,\epsilon_1}} |L_{1,n,y,k}(x) - L_{2,n,y,k}(x)| < \epsilon_4. \quad (53)$$

We then analyze $L_{2,n,y,k}(x)$. Applying a Taylor expansion of $\Phi \left(\frac{y - \eta_{k,n} x}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} \right)$:

$$\begin{aligned}
 \Phi \left(\frac{z - \eta_{k,n} x}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} \right) &= \Phi \left(\frac{Q_{k,n} - x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right) + \phi \left(\frac{Q_{k,n} - x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right) \\
 &\quad \times \left(\frac{z - \eta_{k,n} x}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} - \frac{Q_{k,n} - x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right) \\
 &\quad - \frac{z_x \phi(z_x)}{2} \left(\frac{z - \eta_{k,n} x}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} - \frac{Q_{k,n} - x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right)^2 \\
 &= \Phi \left(\frac{Q_{k,n} - x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right) + \phi \left(\frac{Q_{k,n} - x}{\sqrt{(k-1) \Lambda_n''(\eta_{k,n})}} \right) \\
 &\quad \times \left(\frac{z - \eta_{k,n} Q_{k,n}}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} \right) - \frac{z_x \phi(z_x)}{2} \left(\frac{y - \eta_{k,n} Q_{k,n}}{\sqrt{\frac{k-1}{k}} \psi_{k,n}} \right)^2,
 \end{aligned}$$

for some z_x between $\frac{z-\eta_{k,n}x}{\sqrt{\frac{k-1}{k}}\psi_{k,n}}$ and $\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}}$. Dividing the integral into two parts, we have

$$\begin{aligned}
 L_{2,n,y,k}(x) = & \int_{\eta_{k,n}Q_{k,n}}^T \sqrt{\frac{k-1}{k}}\psi_{k,n}e^{-z} \left(\phi \left(\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}} \right) \left(\frac{z-\eta_{k,n}Q_{k,n}}{\sqrt{\frac{k-1}{k}}\psi_{k,n}} \right) \right. \\
 & + \frac{\alpha_3}{6\sqrt{k-1}} \left(1 - \left(\frac{z-\eta_{k,n}x}{\sqrt{\frac{k-1}{k}}\psi_{k,n}} \right)^2 \right) \phi \left(\frac{z-\eta_{k,n}x}{\sqrt{\frac{k-1}{k}}\psi_{k,n}} \right) \\
 & - \frac{z_x\phi(z_x)}{2} \left(\frac{z-\eta_{k,n}Q_{k,n}}{\sqrt{\frac{k-1}{k}}\psi_{k,n}} \right)^2 - \frac{\alpha_3}{6\sqrt{k-1}} \left(1 - \left(\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}} \right)^2 \right) \\
 & \phi \left(\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}} \right) + \phi \left(\frac{z-\eta_{k,n}x}{\sqrt{\frac{k-1}{k}}\psi_{k,n}} \right) \frac{S_{k-1,n} \left(\frac{z-\eta_{k,n}x}{\sqrt{\frac{k-1}{k}}\psi_{k,n}} \right)}{\sqrt{k-1}} \\
 & - \phi \left(\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}} \right) \frac{S_{k-1,n} \left(\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}} \right)}{\sqrt{k-1}} \Big) dz \\
 & + \int_T^\infty \sqrt{\frac{k-1}{k}}\psi_{k,n}e^{-z} \left(\phi \left(\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}} \right) \left(\frac{z-\eta_{k,n}Q_{k,n}}{\sqrt{\frac{k-1}{k}}\psi_{k,n}} \right) \right. \\
 & + \frac{\alpha_3}{6\sqrt{k-1}} \left(1 - \left(\frac{z-\eta_{k,n}x}{\sqrt{\frac{k-1}{k}}\psi_{k,n}} \right)^2 \right) \phi \left(\frac{z-\eta_{k,n}x}{\sqrt{\frac{k-1}{k}}\psi_{k,n}} \right) \\
 & - \frac{z_x\phi(z_x)}{2} \left(\frac{z-\eta_{k,n}Q_{k,n}}{\sqrt{\frac{k-1}{k}}\psi_{k,n}} \right)^2 - \frac{\alpha_3}{6\sqrt{k-1}} \left(1 - \left(\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}} \right)^2 \right) \\
 & \phi \left(\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}} \right) \\
 & + \phi \left(\frac{z-\eta_{k,n}x}{\sqrt{\frac{k-1}{k}}\psi_{k,n}} \right) \frac{S_{k-1,n} \left(\frac{z-\eta_{k,n}x}{\sqrt{\frac{k-1}{k}}\psi_{k,n}} \right)}{\sqrt{k-1}} \\
 & - \phi \left(\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}} \right) \frac{S_{k-1,n} \left(\frac{Q_{k,n}-x}{\sqrt{(k-1)\Lambda_n''(\eta_{k,n})}} \right)}{\sqrt{k-1}} \Big) dz,
 \end{aligned}$$

where T can be arbitrarily large but fixed. Let $x < V$, where V can be arbitrarily large but fixed. Then by a similar analysis to that of $C_{n,y,k}$, we conclude that for any arbitrarily small but fixed $\epsilon_5 > 0$, there exists a sufficiently large T , such that for

sufficiently small ϵ_1 and sufficiently large n , the following holds for all $x < V$:

$$\sup_{k \in O_{n,y,\epsilon_1}} \left| L_{2,n,y,k}(x) - \frac{1}{\sqrt{2\pi}} e^{-\eta_{k,n} Q_{k,n}} \right| < \epsilon_5. \quad (54)$$

Now we have that

$$\begin{aligned} (52) &= \sqrt{2\pi \frac{k}{k-1}} e^{\Lambda_n^*(Q_{k,n})} \int_0^V x^2 e^{\Lambda_n(\eta_{k,n})} L_{1,n,y,k}(x) d\tilde{v}_{n,y,k}(x) \\ &\quad + \sqrt{2\pi \frac{k}{k-1}} e^{\Lambda_n^*(Q_{k,n})} \int_V^\infty x^2 e^{\Lambda_n(\eta_{k,n})} L_{1,n,y,k}(x) d\tilde{v}_{n,y,k}(x). \end{aligned}$$

Because $L_{1,n,y,k}(x)$ is bounded, $\mathbb{E}_{\tilde{v}_{n,y,k}}[Y_i] = 0$, and $\mathbb{E}_{\tilde{v}_{n,y,k}}[Y_i^2] = 1$, by Lemma 12, we conclude that the second term goes to zero as V goes to infinity uniformly for $k \in O_{n,y,\epsilon_1}$. For the first term, since V can be arbitrarily large, the desired result follows from Lemma 12, (53), and (54). $\square$

8.1.6 Proof of Lemma 13

Proof We first notice that

$$\alpha_{3,n,y,q_{k,n}} = \frac{rf_1(\eta_{k,n})}{\left(\sqrt{\Lambda_n''(\eta_{k,n})}\right)^3} \left(f_4(\eta_{k,n}) - 3f_3^2(\eta_{k,n}) Q_{k,n} + 3g_2(\eta_{k,n}) Q_{k,n}^2 - Q_{k,n}^3 \right),$$

where f_1 through f_4 are defined in (44) through (47). The result for $\alpha_{3,n,y,q_{k,n}}$ then follows from Lemma 12 and the monotonicity and continuity of f_1 through f_4 . Note that a similar argument can be applied to the sixth moment, and thus we can use Holder's inequality to obtain the upper bound for $\beta_{3,n,y,q_{k,n}}$. The lower bound for $\beta_{3,n,y,q_{k,n}}$ follows from Holder's inequality, and the fact that $\mathbb{E}_{\tilde{v}_{n,y,k}}[Y_i^2] = 1$. $\square$

8.1.7 Proof of Lemma 14

Proof The proof follows that of Theorem 1 in §43 of Gnedenko and Kolmogorov (1954). We adopt its structure and notation. Let C be some positive constant whose value might change from line to line. Let

$$G_{k,n}(x) = \Phi(x) + \frac{\alpha_3}{6\sqrt{k}} (1 - x^2) \phi(x) + D_{k,n}(x),$$

and recall that

$$D_{k,n}(x) = \phi(x) \frac{S_{k,n}(x)}{\sqrt{k}}.$$

We can obtain the Fourier-Stieltjes transform of $G_{k,n}$ (see page 214 of Gnedenko and Kolmogorov (1954)):

$$g_{k,n}(t) = e^{-\frac{t^2}{2}} + \frac{\alpha_3(it)^3}{6\sqrt{k}} e^{-\frac{t^2}{2}} + d_{k,n}(t),$$

where

$$\begin{aligned} d_{k,n}(t) &= -\frac{t}{\tau_{k,n}\sqrt{2\pi k}} \sum_{v=-\infty}^{\infty} \frac{1}{v} \int_{-\infty}^{\infty} e^{itx - \frac{x^2}{2} + iv\tau_{k,n}\sqrt{k}(x - x_{k,n}\sqrt{k})} dx \\ &= -\frac{t}{\tau_{k,n}\sqrt{k}} \sum_{v=-\infty}^{\infty} \frac{e^{-i\tau_{k,n}v k x_{k,n}}}{v} e^{-\frac{1}{2}(t + \tau_{k,n}v\sqrt{k})^2}, \end{aligned}$$

and the summation is over every integer $v \neq 0$. It is not hard to observe that

$$\sup_{k \in O_{n,y,\epsilon_1}} |G'_{k,n}(x)|$$

is bounded wherever the derivative exists for sufficiently large n and sufficiently small ϵ_1 . We denote by A its bound. Let $T = n$, then $T \frac{h_{k,n}}{\sqrt{k}}$ (recall that $h_{k,n}$ is of order $O(n^{-\alpha})$) can be arbitrarily large for sufficiently large n (this is required to apply Theorem 2 in §39 of Gnedenko and Kolmogorov (1954)). Let $f_{k,n}(\cdot)$ be the Fourier-Stieltjes transform of $F_{k,n}$. By Theorem 2 in §39 of Gnedenko and Kolmogorov (1954), to prove the desired result, it suffices to show that for sufficiently small ϵ_1 and sufficiently large n

$$\sup_{k \in O_{n,y,\epsilon_1}} \sqrt{k} \int_{-T}^T \left| \frac{f_{k,n}(t) - g_{k,n}(t)}{t} \right| dt \leq \frac{\epsilon}{2}.$$

Following (Gnedenko and Kolmogorov 1954), let

$$\begin{aligned} e_1 &= \int_{-T}^{-\frac{\tau_{k,n}}{2}\sqrt{k}} \left| \frac{f_{k,n}(t) - g_{k,n}(t)}{t} \right| dt, \\ e_2 &= \int_{-\frac{\tau_{k,n}}{2}\sqrt{k}}^{\frac{\tau_{k,n}}{2}\sqrt{k}} \left| \frac{f_{k,n}(t) - g_{k,n}(t)}{t} \right| dt, \text{ and} \\ e_3 &= \int_{\frac{\tau_{k,n}}{2}\sqrt{k}}^T \left| \frac{f_{k,n}(t) - g_{k,n}(t)}{t} \right| dt. \end{aligned}$$

We first analyze e_2 . Let $T_k = \frac{\sqrt{k}}{24\beta_3}$. From Lemma 13, we know that β_3 is a finite positive number bounded away from zero. By the definition of $\tau_{k,n}$, for sufficiently

large n , we have $T_k < \frac{\tau_{k,n}}{2} \sqrt{k}$. Recall that $F_{k,n}$ is the distribution function of

$$k^{-\frac{1}{2}} \sum_{i=1}^k Y_i.$$

Let $\varphi_{k,n}(t)$ be the characteristic function of Y_i when $e^{-\lambda_1(\zeta_n - y)} X_i$ has distribution $\tilde{\nu}_{n,y,k}$. We have the following result which gives an upper bound for $|\varphi_{k,n}(t)|$.

Lemma 16 *There exists $c_1 > 0$ such that for sufficiently large n and sufficiently small ϵ_1 , when $T_k/\sqrt{k} \leq |t| \leq \tau_{k,n}/2$,*

$$\sup_{k \in O_{n,y,\epsilon_1}} |\varphi_{k,n}(t)| < e^{-c_1}.$$

Proof See Section 8.1.9. □

For sufficiently large n , by Lemma 16, the following holds for $T_k \leq |t| \leq \tau_{k,n}\sqrt{k}/2$:

$$|f_{k,n}(t)| = \left| \varphi_{k,n}\left(\frac{t}{\sqrt{k}}\right) \right|^k < e^{-c_1 k}.$$

We also note that for some $c_2 > 0$ and sufficiently large n and $k \in O_{n,y,\epsilon_1}$, the following holds when $T_k \leq |t| \leq \tau_{k,n}\sqrt{k}/2$:

$$|g_{k,n}(t)| \leq e^{-c_2 k}.$$

Therefore, for some positive constant c ,

$$e_2 \leq \int_{-T_k}^{T_k} \left| \frac{f_{k,n}(t) - g_{k,n}(t)}{t} \right| dt + 4 \int_{T_k}^{\frac{1}{2}\tau_{k,n}\sqrt{k}} \frac{e^{-ck}}{t} dt.$$

By Theorem 1(b) in §41 of Gnedenko and Kolmogorov (1954), there exists some positive constant C such that for sufficiently large n and $k \in O_{n,y,\epsilon_1}$,

$$\int_{-T_k}^{T_k} \left| \frac{f_{k,n}(t) - g_{k,n}(t)}{t} \right| dt \leq \frac{C}{\sqrt{k}} \delta(k) + \int_{-T_k}^{T_k} \left| \frac{d_{k,n}(t)}{t} \right| dt,$$

where $\delta(k)$ only depends on k , and $\lim_{k \rightarrow \infty} \delta(k) = 0$. We also notice that for $|t| \leq T_k$, there exists some positive constant C such that for sufficiently large n and $k \in O_{n,y,\epsilon_1}$,

$$\left| \frac{d_{k,n}(t)}{t} \right| \leq \frac{C}{\tau_{k,n}\sqrt{k}} e^{-\frac{t^2}{2}}.$$

Therefore, for sufficiently large n and $k \in O_{n,y,\epsilon_1}$,

$$e_2 \leq C \left(\frac{\delta(k)}{\sqrt{k}} + \log(n) e^{-ck} + \frac{1}{\tau_{k,n} \sqrt{k}} \right),$$

which is of order $o\left(\frac{1}{\sqrt{k}}\right)$.

We then estimate e_3 . For sufficiently large n and $k \in O_{n,y,\epsilon_1}$,

$$\begin{aligned} e_3 &= \int_{\frac{\tau_{k,n}}{2}\sqrt{k}}^n \left| \frac{f_{k,n}(t) - g_{k,n}(t)}{t} \right| dt \\ &\leq \frac{C}{k} + \int_{\frac{\tau_{k,n}}{2}\sqrt{k}}^n \left| \frac{f_{k,n}(t) - d_{k,n}(t)}{t} \right| dt \quad (\text{Mill's Inequality}) \\ &= \frac{C}{k} + \int_{\frac{\tau_{k,n}}{2}}^{n/\sqrt{k}} \left| \frac{\varphi_{k,n}^k(t) - d_{k,n}(\sqrt{kt})}{t} \right| dt \\ &= \frac{C}{k} + \sum_{j=1}^J \int_{\frac{2j-1}{2}\tau_{k,n}}^{\frac{2j+1}{2}\tau_{k,n}} \left| \frac{\varphi_{k,n}^k(t) - d_{k,n}(\sqrt{kt})}{t} \right| dt \\ &\quad + \int_{\frac{2J+1}{2}\tau_{k,n}}^{n/\sqrt{k}} \left| \frac{\varphi_{k,n}^k(t) - d_{k,n}(\sqrt{kt})}{t} \right| dt, \end{aligned}$$

where

$$J = \left\lceil \frac{n/\sqrt{k}}{\tau_{k,n}} - \frac{1}{2} \right\rceil.$$

Let

$$I_j = \int_{\frac{2j-1}{2}\tau_{k,n}}^{\frac{2j+1}{2}\tau_{k,n}} \left| \frac{\varphi_{k,n}^k(t) - d_{k,n}(\sqrt{kt})}{t} \right| dt.$$

Apply a change of variable $t = z + j\tau_{k,n}$, we have

$$\begin{aligned} I_j &= \int_{-\frac{\tau_{k,n}}{2}}^{\frac{\tau_{k,n}}{2}} \left| \frac{e^{ij\tau_{k,n}x_{k,n}k} \varphi_{k,n}^k(z) + \frac{z+j\tau_{k,n}}{\tau_{k,n}} \sum_v \frac{1}{v} e^{-iv\tau_{k,n}x_{k,n}k} e^{-\frac{k}{2}(z+j\tau_{k,n}+v\tau_{k,n})^2}}{z+j\tau_{k,n}} \right| dz \\ &\leq \int_{-\frac{\tau_{k,n}}{2}}^{\frac{\tau_{k,n}}{2}} \left| \frac{\varphi_{k,n}^k(z) - e^{-\frac{1}{2}kz^2} - \frac{z}{j\tau_{k,n}} e^{-\frac{1}{2}kz^2}}{z+j\tau_{k,n}} \right| dz + C e^{-\frac{k\tau_{k,n}^2}{8}}. \end{aligned}$$

where the inequality is due to the fact that the term in the inner summation only makes considerable contribution to the integral when $v = -j$ (note that when $v \neq -j$,

$$e^{-\frac{k}{2}(z+j\tau_{k,n}+v\tau_{k,n})^2} = O\left(e^{-\frac{k\tau_{k,n}^2}{8}}\right)$$

uniformly in z ; also see page 216 of Gnedenko and Kolmogorov (1954)). We first note that

$$\int_{-\frac{\tau_{k,n}}{2}}^{\frac{\tau_{k,n}}{2}} \left| \frac{\varphi_{k,n}^k(z) - e^{-\frac{1}{2}kz^2}}{z + j\tau_{k,n}} \right| dz \leq \frac{2}{j\tau_{k,n}} \int_{-\frac{\tau_{k,n}}{2}\sqrt{k}}^{\frac{\tau_{k,n}}{2}\sqrt{k}} \left| f_{k,n}(z) - e^{-\frac{1}{2}z^2} \right| \frac{dz}{\sqrt{k}}.$$

By a similar argument to the treatment of e_2 , we can obtain that for sufficiently large n and $k \in O_{n,y,\epsilon_1}$,

$$\int_{-\frac{\tau_{k,n}}{2}}^{\frac{\tau_{k,n}}{2}} \left| \frac{\varphi_{k,n}^k(z) - e^{-\frac{1}{2}kz^2}}{z + j\tau_{k,n}} \right| dz \leq \frac{C}{jk}.$$

We then note that for sufficiently large n and $k \in O_{n,y,\epsilon_1}$,

$$\frac{1}{j\tau_{k,n}} \int_{-\frac{\tau_{k,n}}{2}}^{\frac{\tau_{k,n}}{2}} \left| \frac{z}{z + j\tau_{k,n}} \right| e^{-\frac{1}{2}kz^2} dz \leq \frac{C}{j^2k}.$$

Recall that $J = \left[\frac{n/\sqrt{k}}{\tau_{k,n}} - \frac{1}{2} \right]$. For sufficiently large n , we have $e_3 \leq C \log(n)/k$. By the same method, $e_1 \leq C \log(n)/k$. Therefore, for sufficiently large n and $k \in O_{n,y,\epsilon_1}$

$$\int_{-T}^T \left| \frac{f_{k,n}(t) - g_{k,n}(t)}{t} \right| dt \leq C \left(\frac{\delta(k)}{\sqrt{k}} + \log(n) e^{-ck} + \frac{1}{\tau_{k,n}\sqrt{k}} + \frac{\log(n)}{k} \right).$$

An application of Theorem 2 of §39 of Gnedenko and Kolmogorov (1954) leads to the desired result. $\square$

8.1.8 Proof of Lemma 15

Proof Following the proof of Theorem 3.7.4 in Dembo and Zeitouni (2009), we have

$$\mathbb{P} \left(\sum_{i=1}^k e^{-\lambda_1(\zeta_n - y)} X_i \geq k Q_{k,n} \right) J_{n,y,k} = \sqrt{2\pi} \int_0^\infty \psi_{k,n} e^{-t} \left[F_{k,n} \left(\frac{t}{\psi_{k,n}} \right) - F_{k,n}(0) \right].$$

The desired result follows directly by Lemma 12, Lemma 13 and Lemma 14. $\square$

8.1.9 Proof of Lemma 16

Proof

$$\begin{aligned}
 \varphi_{k,n}(t) &= \mathbb{E}_{\tilde{v}_{n,y,k}} \left[e^{it \frac{e^{-\lambda_1(\zeta_n-y)} X_1 - Q_{k,n}}{\sqrt{\Lambda_n''(\eta_{k,n})}}} \right] \\
 &= \mathbb{E} \left[e^{it \frac{e^{-\lambda_1(\zeta_n-y)} X_1 - Q_{k,n}}{\sqrt{\Lambda_n''(\eta_{k,n})}}} e^{\eta_{k,n} e^{-\lambda_1(\zeta_n-y)} X_1 - \Lambda_n(\eta_{k,n})} \right] \\
 &= e^{-it \frac{Q_{k,n}}{\sqrt{\Lambda_n''(\eta_{k,n})}}} e^{-\Lambda_n(\eta_{k,n})} \mathbb{E} \left[e^{it \frac{e^{-\lambda_1(\zeta_n-y)} X_1}{\sqrt{\Lambda_n''(\eta_{k,n})}}} e^{\eta_{k,n} e^{-\lambda_1(\zeta_n-y)} X_1} \right] \\
 &= e^{-it \frac{Q_{k,n}}{\sqrt{\Lambda_n''(\eta_{k,n})}}} e^{-\Lambda_n(\eta_{k,n})} \frac{\int_0^{\zeta_n-y} e^{\lambda_0 s} \frac{e^{-\lambda_1(\zeta_n-y-s)} e^{f(t) e^{-\lambda_1(\zeta_n-y)}}}{e^{-\lambda_1(\zeta_n-y-s)} e^{f(t) e^{-\lambda_1(\zeta_n-y)}} - e^{f(t) e^{-\lambda_1(\zeta_n-y)}} + 1} ds}{\int_0^{\zeta_n-y} e^{\lambda_0 s} ds} \\
 &= e^{-it \frac{Q_{k,n}}{\sqrt{\Lambda_n''(\eta_{k,n})}}} \frac{\int_0^{\zeta_n-y} e^{\lambda_0 s} \frac{e^{-\lambda_1(\zeta_n-y-s)} e^{f(t) e^{-\lambda_1(\zeta_n-y)}}}{e^{-\lambda_1(\zeta_n-y-s)} e^{f(t) e^{-\lambda_1(\zeta_n-y)}} - e^{f(t) e^{-\lambda_1(\zeta_n-y)}} + 1} ds}{\int_0^{\zeta_n-y} e^{\lambda_0 s} \frac{e^{-\lambda_1(\zeta_n-y-s)} e^{e^{-\lambda_1(\zeta_n-y)} \eta_{k,n}}}{e^{-\lambda_1(\zeta_n-y-s)} e^{e^{-\lambda_1(\zeta_n-y)} \eta_{k,n}} - e^{e^{-\lambda_1(\zeta_n-y)} \eta_{k,n}} + 1} ds},
 \end{aligned}$$

where $f(t) = \frac{it}{\sqrt{\Lambda_n''(\eta_{k,n})}} + \eta_{k,n}$. Let

$$A_{n,s} = e^{-\lambda_1(\zeta_n-y-s)},$$

$$B_{k,n} = e^{e^{-\lambda_1(\zeta_n-y)} \eta_{k,n}},$$

$$C_{k,n,t} = \cos \left(\frac{t}{\sqrt{\Lambda_n''(\eta_{k,n})}} e^{-\lambda_1(\zeta_n-y)} \right) - 1.$$

For simplicity, we omit all subscripts and refer to them as A , B and C . For some positive T ,

$$\begin{aligned}
 |\varphi_{k,n}(t)| &\leq \frac{\int_0^{\zeta_n-y} e^{\lambda_0 s} \frac{AB}{\sqrt{(AB-B+1)^2 + 2(AB-B)C}} ds}{\int_0^{\zeta_n-y} e^{\lambda_0 s} \frac{AB}{AB-B+1} ds} \\
 &= \frac{\int_0^T e^{\lambda_0 s} \frac{AB}{\sqrt{(AB-B+1)^2 + 2(AB-B)C}} ds + \int_T^{\zeta_n-y} e^{\lambda_0 s} \frac{AB}{\sqrt{(AB-B+1)^2 + 2(AB-B)C}} ds}{\int_0^T e^{\lambda_0 s} \frac{AB}{AB-B+1} ds + \int_T^{\zeta_n-y} e^{\lambda_0 s} \frac{AB}{AB-B+1} ds} \\
 &< \frac{\int_0^T e^{\lambda_0 s} \frac{AB}{\sqrt{(AB-B+1)^2 + 2(AB-B)C}} ds + \int_T^{\zeta_n-y} e^{\lambda_0 s} \frac{AB}{AB-B+1} ds}{\int_0^T e^{\lambda_0 s} \frac{AB}{AB-B+1} ds + \int_T^{\zeta_n-y} e^{\lambda_0 s} \frac{AB}{AB-B+1} ds}.
 \end{aligned}$$

We notice that for $T_k/\sqrt{k} \leq |t| \leq \tau_{k,n}/2$, the final term achieves its maximum value when $t = T_k/\sqrt{k}$ ($|C|$ is minimized at $t = T_k/\sqrt{k}$). By Lemma 12, let η_1 and η_2 be the lower bound and upper bound for $\eta_{k,n}$ when n is sufficiently large and ϵ_1 is sufficiently small. By checking the sign of derivative with respect to $\eta_{k,n}$ and n (with the help of Leibniz integral rule), we can obtain that

$$\int_0^{\zeta_n-y} e^{\lambda_0 s} \frac{AB}{AB-B+1} ds \leq \int_0^\infty e^{\lambda_0 s} \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \eta_2} ds. \quad (55)$$

Moreover, for fixed t ,

$$\lim_{n \rightarrow \infty} \frac{1 - \cos\left(\frac{t}{\sqrt{\Lambda_n''(\eta)}} e^{-\lambda_1(\zeta_n-y)}\right)}{e^{-2\lambda_1(\zeta_n-y)}} = \lim_{n \rightarrow \infty} \frac{t^2}{2\Lambda_n''(\eta)} = \frac{t^2}{2\Lambda''(\eta)},$$

where

$$\Lambda''(\eta) = \frac{2 \int_0^\infty e^{\lambda_0 s} \frac{e^{-2\lambda_1 s}}{(1-\eta e^{-\lambda_1 s})^3} ds \int_0^\infty e^{\lambda_0 s} \frac{1}{1-\eta e^{-\lambda_1 s}} ds - \left(\int_0^\infty e^{\lambda_0 s} \frac{e^{-\lambda_1 s}}{(1-\eta e^{-\lambda_1 s})^2} ds \right)^2}{\left(\int_0^\infty e^{\lambda_0 s} \frac{1}{1-\eta e^{-\lambda_1 s}} ds \right)^2}.$$

Hence,

$$\begin{aligned} |\varphi_{k,n}(t)| &< 1 + \frac{\int_0^T e^{\lambda_0 s} \frac{AB}{\sqrt{(AB-B+1)^2 + 2(AB-B)C}} ds - \int_0^T e^{\lambda_0 s} \frac{AB}{AB-B+1} ds}{\int_0^T e^{\lambda_0 s} \frac{AB}{AB-B+1} ds + \int_0^{\zeta_n-y} e^{\lambda_0 s} \frac{AB}{AB-B+1} ds} \\ &= 1 + \frac{\int_0^T e^{\lambda_0 s} \frac{AB}{\sqrt{(AB-B+1)^2 + 2(AB-B)C}} ds - \int_0^T e^{\lambda_0 s} \frac{AB}{AB-B+1} ds}{\int_0^{\zeta_n-y} e^{\lambda_0 s} \frac{AB}{AB-B+1} ds}. \end{aligned}$$

We notice that

$$\begin{aligned} &\limsup_{n \rightarrow \infty} \sup_{\eta_{k,n} \in [\eta_1, \eta_2]} \int_0^T e^{\lambda_0 s} \frac{AB}{\sqrt{(AB-B+1)^2 + 2(AB-B)C}} ds \\ &\quad - \int_0^T e^{\lambda_0 s} \frac{AB}{AB-B+1} ds \\ &\geq \sup_{\eta_{k,n} \in [\eta_1, \eta_2]} \lim_{n \rightarrow \infty} \int_0^T e^{\lambda_0 s} \frac{AB}{\sqrt{(AB-B+1)^2 + 2(AB-B)C}} ds \\ &\quad - \int_0^T e^{\lambda_0 s} \frac{AB}{AB-B+1} ds \\ &= \sup_{\eta \in [\eta_1, \eta_2]} \int_0^T e^{\lambda_0 s} \frac{e^{\lambda_1 s}}{\sqrt{(e^{\lambda_1 s} - \eta)^2 + \frac{(T_k/\sqrt{k})^2}{\Lambda''(\eta)}}} ds - \int_0^T e^{\lambda_0 s} \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \eta} ds, \end{aligned}$$

where the last line is less than zero and only depends on β_3 by recalling that $T_k = \frac{\sqrt{k}}{24\beta_3}$. From Lemma 13, we know that β_3 is bounded for sufficiently large n and sufficiently small ϵ_1 . Therefore, for any $\epsilon > 0$, by (55), we have

$$\begin{aligned} & 1 + \frac{\int_0^T e^{\lambda_0 s} \frac{AB}{\sqrt{(AB-B+1)^2 + 2(AB-B)C}} ds - \int_0^T e^{\lambda_0 s} \frac{AB}{AB-B+1} ds}{\int_0^{\zeta_n - y} e^{\lambda_0 s} \frac{AB}{AB-B+1} ds} \\ & \leq 1 + \sup_{\eta \in [\eta_1, \eta_2]} \frac{\int_0^T e^{\lambda_0 s} \frac{e^{\lambda_1 s}}{\sqrt{(e^{\lambda_1 s} - \eta)^2 + \frac{(T_k/\sqrt{k})^2}{\Lambda''(\eta)}}} ds - \int_0^T e^{\lambda_0 s} \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \eta} ds}{\int_0^\infty e^{\lambda_0 s} \frac{e^{\lambda_1 s}}{e^{\lambda_1 s} - \eta_1} ds} + \epsilon, \end{aligned}$$

for sufficiently large n and sufficiently small ϵ_1 . The desired result then follows. $\square$

Acknowledgements The authors would like to thank Xuanming Zhang for helpful comments on the draft. The work of KL was partially supported by NSF award CMMI 2228034. The work of ZW was supported by the Guangdong (China) Provincial Key Laboratory of Mathematical Foundations for Artificial Intelligence [2023B1212010001].

Declarations

Conflicts of Interest None.

References

- Athreya KB, Ney PE (1972) Branching processes. Springer & Verlag, Berlin
- Avanzini S, Antal T (2019) Cancer recurrence times from a branching process model. *PLoS Comput Biol* 15(11):e1007423
- Border KC (2020) Miscellaneous notes on optimization theory and related topics. <http://www.its.caltech.edu/~kcborder/Notes/Maximization.pdf>
- Burgess MR, Sawyers CL et al (2006) Treating imatinib-resistant leukemia: the next generation targeted therapies. *Sci World J* 6:918–930
- Cheek D, Antal T (2018) Mutation frequencies in a birth–death branching process. *Ann Appl Probab* 28(6):3922–3947
- Coldman AJ, Goldie JH (1986) A stochastic model for the origin and treatment of tumors containing drug-resistant cells. *Bull Math Biol* 48:279–292
- Dagogo-Jack I, Shaw AT (2018) Tumour heterogeneity and resistance to cancer therapies. *Nat Rev Clin Oncol* 15(2):81–94
- Dembo A, Zeitouni O (2009) Large deviations techniques and applications. Springer, Berlin
- den Hollander F (2000) Large Deviations. American Mathematical Society
- Durrett R (2014) Branching process models of cancer. Springer International Publishing, Berlin
- Durrett R, Moseley S (2009) Evolution of resistance and progression to disease during clonal expansion of cancer. *Theor Popul Biol* 77(1):42–48
- Durrett R, Foo J, Leder K, Mayberry J, Michor F (2011) Intratumor heterogeneity in evolutionary models of tumor progression. *Genetics* 188(2):461–477
- Esseen C-G (1945) Fourier analysis of distribution functions. A mathematical study of the laplace-gaussian law. *Acta Math* 77:1–125
- Foo J, Leder K (2013) Dynamics of cancer recurrence. *Ann Appl Probab* 23(4):1437–1468
- Foo J, Leder K, Zhu J (2014) Escape times for branching processes with random mutational fitness effects. *Stoch Proc Appl* 124(11):3661–3697
- Gambacorti-Passerini C et al (2011) Multicenter independent assessment of outcomes in chronic myeloid leukemia patients treated with imatinib. *J Natl Cancer Inst* 103(7):553–561

- Gnedenko BV, Kolmogorov AN (1954) Limit distributions for sums of independent random variables. Addison-Wesley, Boston
- Haeno H, Iwasa Y, Michor F (2009) The evolution of two mutations during clonal expansion. *Genetics* 177(4):2209–2221
- Hanagal P, Leder K, Wang Z (2022) Large deviations of cancer recurrence timing. *Stoch Proc Appl* 147:1–50
- Hunter C et al (2006) A hypermutation phenotype and somatic msh6 mutations in recurrent human malignant gliomas after alkylator chemotherapy. *Can Res* 66(8):3987–3991
- Iwasa Y, Michor F, Nowak M (2003) Evolutionary dynamics of escape from biomedical intervention. *Proc Biol Sci* 270:2572–2578
- Jagers P, Klebaner FC, Sagitov S (2007) Markovian paths to extinction. *Adv Appl Probab* 39:569–587
- Jagers P, Klebaner FC, Sagitov S (2007) On the path to extinction. *Proc Natl Acad Sci* 104:6107–6111
- Keller P, Antal T (2015) Mutant number distribution in an exponentially growing population. *J Stat Mech: Theory Exp* 2015(1):P01011
- Komarova N (2006) Stochastic modeling of drug resistance in cancer. *J Theoret Biol* 239:351–366
- Leder K, Sun R, Wang Z, Zhang X (2024) Parameter estimation from single patient, single time-point sequencing data of recurrent tumors. *J Math Biol* 89(5):51
- Leder K, Wang Z, Zhang X (2025) Parameter estimation in recurrent tumor evolution with finite carrying capacity. [arXiv:2510.01078](https://arxiv.org/abs/2510.01078)
- Michelson S, Ito K, Tran HT, Leith JT (1989) Stochastic models for subpopulation emergence in heterogeneous tumors. *Bull Math Biol* 51(6):731–747
- Misale S, Yaeger R, Hobor S, Scala E, Janakiraman M, Liska D, Valtorta E, Schiavo R, Buscarino M, Siravegna G et al (2012) Emergence of kras mutations and acquired resistance to anti-egfr therapy in colorectal cancer. *Nature* 486(7404):532–536
- Sagitov S, Serra M (2009) Multitype Bienayme-Galton-Watson processes escaping extinction. *Adv Appl Probab* 41:225–246
- Sehl M, Zhou H, Sinsheimer J, Kenneth L (2011) Extinction models for cancer stem cell therapy. *Math Biosci* 234:132–146
- Wang X, Zhang H, Chen X (2019) Drug resistance and combating drug resistance in cancer. *Cancer Drug Resist* 2(2):141

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.